

Modelling the Mediterranean Sea ecosystem at high spatial resolution to inform the ecosystem-based management in the region

Chiara Piroddi, Marta Coll, Diego Macias Moy, Jeroen Steenbeek, Elisa Garcia-Gorriz, Alessandro Mannini, Daniel Vilas, Villy Christensen

S1.

Ecopath baseline model

Ecopath is a mass-balanced model based on the assumption that the production of one functional group is equal to the sum of all predation, non-predatory losses, exports, biomass accumulations, and catches, as expressed by the following equation:

$$P/B_i \cdot B_i = P/B_i \cdot B_i \cdot (1 - EE_i) + \sum_j (Q/B)_{ji} \cdot B_i \cdot DC_{ji} + Y_i + NM_i + BA_i$$

where B_i is the biomass, $(P/B)_i$ is the production rate, $(Q/B)_i$ is the consumption rate, DC_{ji} is the fraction of prey i included in the diet of predator j , NM_i is the net migration of prey i , BA_i is the biomass accumulation of prey i , Y_i is the catch of prey i , and EE_i is the ecotrophic efficiency of prey i , that is, the proportion of production used in the system.

In an Ecopath model, the energy input and output of all functional groups must be balanced under some ecological and thermodynamic rules: (1) $EE < 1.0$; (2) P/Q [production/consumption rate or gross efficiency (GE)] ranges from 0.1 to 0.3 with the exception of fast-growing groups such as bacteria; (3) R/A (respiration/food assimilation) < 1 ; (4) R/B (respiration/biomass) ranges from 1 to 10 for fishes and higher values for small organisms; (5) NE (net efficiency of food conversion) $> GE$ and (6) P/R (production/respiration) < 1 ^{1,2}. To balance the Mediterranean model, we applied a manual mass-balanced procedure following a top-down approach modifying appropriate input parameters (starting from the functional groups with higher trophic levels) and following the best practice guidelines provided in the literature²⁻⁴. Appendix S2 shows the balanced model and its main input parameters.

Ecosim, temporal model

Ecosim model describes the temporal dynamics of species biomass and flows over time by accounting for changes in predation, consumption rate, fishing, and the environment ^{1,5}. Ecosim uses a set of differential equations to describe biomass dynamics:

$$\frac{dB_i}{dt} = \left(\frac{P}{Q}\right)_i \cdot \sum Q_{ji} - \sum Q_{ji} + I_i - (M_i + F_i - e_i) \cdot B_i$$

where dB_i/dt is the growth rate of group i during time t in terms of its biomass B_i ; $(P/Q)_i$ is the net growth efficiency of group i ; Q_{ij} is the consumption rate; M_i is the non-predation mortality rate; F_i is the fishing mortality rate; e_i is the emigration; and I_i is the immigration rate ¹.

Consumption rates (Q_{ij}) in Ecosim are calculated based on the “foraging arena” theory, which divides the biomass of the prey into a vulnerable and a non-vulnerable fraction and the transfer rate or vulnerability between the two fractions determines the trophic flow between the predator and the prey [6]. The vulnerability concept incorporates density-dependence processes and expresses how far a group is from its carrying capacity ¹. Default values of vulnerability ($v = 2$) represent a mixed trophic flow, a low value ($v < 2$) indicates ‘bottom-up’ flow control and a situation closer to carrying capacity, while a high value ($v > 2$) indicates ‘top-down’ flow control and a situation further away from carrying capacity ⁶. For each predator-prey interaction, consumption rates are calculated as:

$$Q_{ij} = \frac{a_{ij} \cdot v_{ij} \cdot B_i \cdot B_j \cdot T_i \cdot T_j \cdot M_{ij} / D_j}{v_{ij} + v_{ij} \cdot T_i \cdot M_{ij} + a_{ij} \cdot M_{ij} \cdot B_j \cdot T_j / D_j} \cdot f(Env_{function}, t),$$

where a_{ij} is the rate of effective search for i by j ; T_i represents prey relative feeding time; T_j the predator relative feeding time; M_{ij} is the mediation forcing effects; v_{ij} is the vulnerability parameter; D_j represents the effects of handling time as a limit to consumption rate; and $f(Env_{function}, t)$ is the environmental response function that restricts the size of the foraging arena (C_{rcj}) to account for external environmental drivers changing over time, such as temperature ^{6,7}.

The Ecosim approach was used here to fit the model to observed time-series of data using the sum of squares (SS) ratio between predicted and observed data (survey biomasses and catches for those functional groups with available information) as a metric for assessing model performance. Fishing effort, primary production (PP) of large and small phytoplankton groups and temperature over time were used as main forcing time series to drive the model.

The fitting procedure followed same approach as the one described and applied by Mackinson ⁸. This method uses the Akaike Information criterion (AIC) ^{9,10}:

$$AIC = n \log (minSS/n) + 2k$$

where n is number of observations, minSS is the minimum sum of squares resulting from the comparison of predicted with observed datasets, and k is the number of parameters, to test statistical hypotheses related to changes in predator-prey dynamics; changes in productivity; impact of fishing and possible combinations of the above-mentioned factors. The AIC is a tool used for model selection that penalizes for fitting too many parameters, and which is used to choose the “best” model (the one yielding the lowest AIC) considering a good fit and the least number of estimated parameters to do so. In this study, we used the second-order Akaike Information Criterion (AICc) calculated as follow:

$$AICc = AIC + 2k(k-1)/(n-k-1)$$

to account for small sample sizes (n of observations) in the dataset. The most statistically significant results in our model fitting exercise were obtained when trophic interactions, fishing and environmental changes (PP and temperature) were included together in the model run (Table 1).

Table S1. Results of the temporally dynamic fitting procedure in Ecosim. Years= number of years in the model; n= number of observation; minSS= minimum sum of squares; K= number of parameters; AICc= second-order Akaike Information Criterion; %IF= improved fit compared to the baseline.

AIC test	no. of years	n (number of observations)	K	minSS	AICc	%IF
Baseline	22	1804	0	883.65	-1287.5	
Fishing + Trophic interactions + Environment	22	1804	43	605.00	-1882.9	46.24

After the temporal dynamic fitting procedure, with the “best” fitted model we run the Monte Carlo routine built into EwE to assess sensitivity of Ecosim’s output to the basic Ecopath input parameters (B, P/B, Q/B, EE, diet). This routine draw input parameters from a normal distribution centered on the base Ecopath value and using a defined coefficient of variation, in this case set to 0.1 ⁴. Here, we run 1000 iterations, and the range of outputs (the 5th and 95th percentile) were plotted for the fitted results (in our case time series of biomasses). Figure 1 shows the modelling fitting results for the functional groups biomasses of the Mediterranean Sea for the 1995-2016 period.

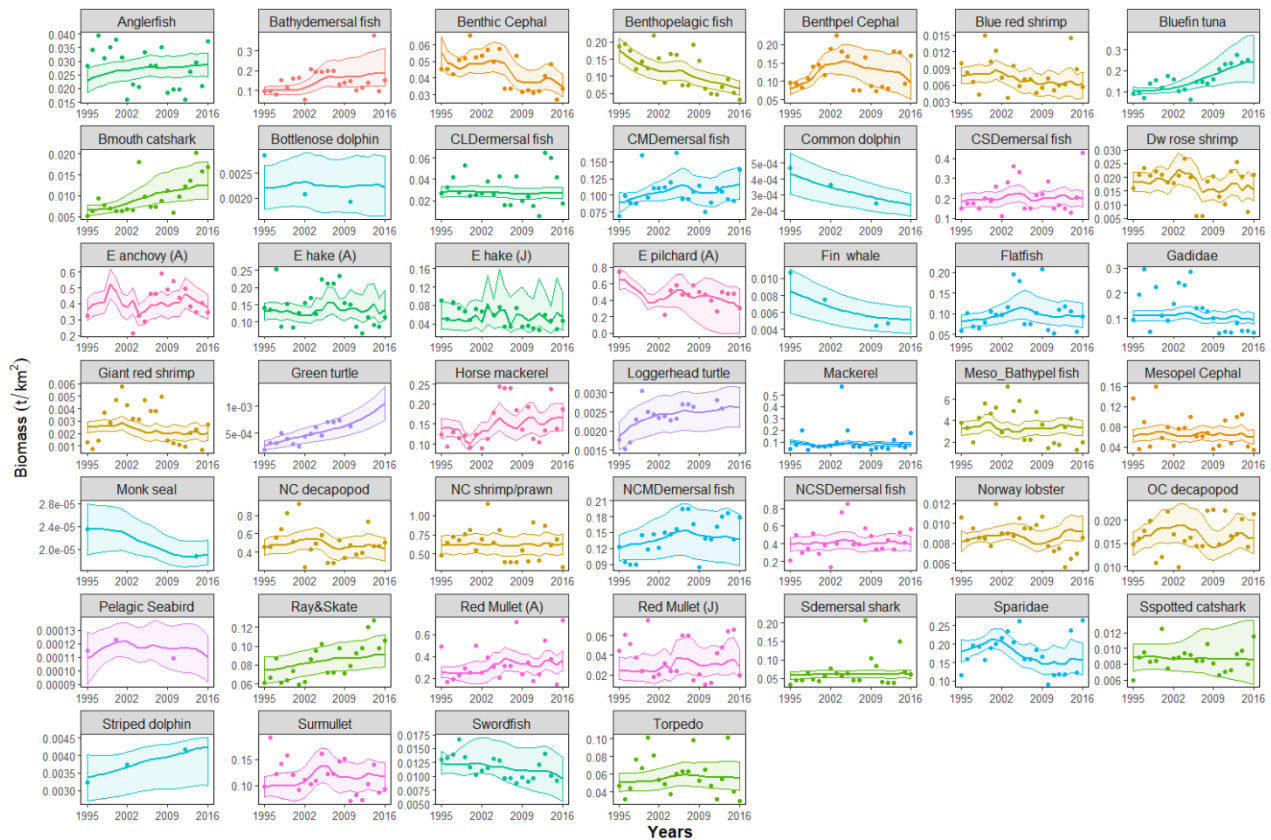


Fig S1. Representation of modelling fitting results for some functional groups occurring in the Mediterranean Sea for the period 1995–2016.

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S2.

Table S.2 Mediterranean functional groups species composition, methods, and references used to estimate the basic input parameters (Biomass (B), Production over Biomass (P/B), Consumption over Biomass (Q/B), Diet (D), By-Catch (B-C), Catch (C), Discard (Di) of the 1990s Ecopath model. Biomass references include also time series data incorporated in Ecosim (1990-2014) module. Abbreviations: E (energy required); W (weight); a (coefficient); z (total mortality); F (fishing mortality); M (natural mortality); T (water temperature); A (aspect ratio caudal fin); h (dummy variable= 1 for herbivores and 0 for carnivores and detritivores); d (dummy variable= 1 for detritivores and 0 for herbivores and carnivores); K (growth coefficient); BT (bottom temperature); D (water depth); Me (dummy variable= 1 for motile epifauna and 0 for others); Car (dummy variable= 1 for carnivorous and 0 for others); Mol (dummy variable= 1 for mollusca and 0 for others); Crus (dummy variable= 1 for crustaceans and 0 for others); Pol (dummy variable= 1 for polychaeta and 0 for others); Echi (dummy variable= 1 for echinodermata and 0 for others)		
Input parameters	Method	Source
1. Bottlenose dolphin: <i>Tursiops truncatus</i>		
B		1-27
P/B	Life history table	28
Q/B	From modified energy requirement equation: $E = aW^{0.714}$	29-33
D		33-37
B-C		38-46
2. Striped dolphin: <i>Stenella coeruleoalba</i>		
B		6,8,10,18,20,21,27,47-57
P/B	Life history table	28
Q/B	From modified energy requirement equation: $E = aW^{0.714}$	29-31,58,59
D		34,37,50,60-66
B-C		38-43,67-70
3. Short-beaked common dolphin: <i>Delphinus delphis</i>		
B		6,21,27,52,71,72
P/B	Life history table	28
Q/B	From modified energy requirement equation: $E = aW^{0.714}$	29-31,73
D		37,71,74,75
B-C		38-42,68,69
4. Fin whale: <i>Balaenoptera physalus</i>		
B		10,47,49-51,56,76-82
P/B	Life history table	28
Q/B	From modified energy requirement equation: $E = aW^{0.714}$	29-31
D		50,83-86

B-C		38-41
5. Sperm whale: <i>Physeter macrocephalus</i>		
B		10,50,87-94
P/B	Life history table	28
Q/B	From modified energy requirement equation: $E = aW^{0.714}$	29-31
D		50,93,95-97
B-C		38-41,68,98
6. Deep sea-cetacean feeder: <i>Globicephala melas</i>, <i>Grampus griseus</i>, <i>Ziphius cavirostris</i>		
B		6,8,10,18,20,50,87,88,99-107
P/B	Life history table	28
Q/B	From modified energy requirement equation: $E = aW^{0.714}$	29-31
D		34,37,50,96,99,108-112
B-C		38-41,67,68
7. Monk seal: <i>Monachus monachus</i>		
B		113-123
P/B	Life history table	28
Q/B	From modified energy requirement equation: $E = aW^{0.75}$	29,30,124
D		125-128
B-C		38,40,41,129
8. Endangered and pelagic seabird: <i>Calonectris diomedea</i>, <i>Hydrobates pelagicus melitensis</i>, <i>Puffinus yelkouan</i>, <i>Puffinus mauretanicus</i>		
B		130-161
P/B	Mortality (z)= -ln(Survival rate)	162
Q/B	Yearly food intake (including breeding and no breeding seasons)/B	163,164
D		130,136,138,155,158,159,165-168
B-C		38,41,140,142,143,153,169,170
9. Gull and cormorant: <i>Larus audouinii</i>, <i>Larus genei</i>, <i>Larus melanocephalus</i>, <i>Larus michahellis</i>, <i>Larus ridibundus</i>, <i>Phalacrocorax aristotelis</i>, <i>Phalacrocorax carbo</i>		
B		130,134,141,142,146-148,151-156,158,160,161,168,171-206
P/B	Mortality (z)= -ln(Survival rate)	162,207,208
Q/B	Yearly food intake (including breeding and no breeding seasons)/B	163,164
D		130,146,155,158,168,173,197,202,209-216
B-C		38,41,142,153,170
10. Tern: <i>Sterna albifrons</i>, <i>Sterna caspia</i>, <i>Sterna hirundo</i>, <i>Sterna nilotica</i>, <i>Sterna sandvicensis</i>		
B		130,146-148,151,154,156,158,160,180,188-190,198,199,201,205,206,217-222
P/B	Mortality (z)= -ln(Survival rate)	162
Q/B	Yearly food intake (including breeding and no breeding seasons)/B	163,164
D		130,155,168,217
B-C		38,41,153

11. Loggerhead turtle: <i>Caretta caretta</i>		
B		223-238
P/B	Mortality (z)= -ln(Survival rate)	162,236,239
Q/B	Yearly food intake/B	240-242
D		243-245
B-C		41,67,69,70,236,246-251
12. Green turtle: <i>Chelonia mydas</i>		
B		223,224,230-232,236,252-254
P/B	Mortality (z)= -ln(Survival rate)	162,236
Q/B	Yearly food intake/B	241,242,255,256
D		255,257
B-C		41,236,249,251
13. Pelagic shark: <i>Alopias vulpinus</i> , <i>Carcharhinus plumbeus</i> , <i>Carcharia taurus</i> , <i>Carcharodon carcharias</i> , <i>Galeorhinus galeus</i> , <i>Isurus oxyrinchus</i> , <i>Lamna nasus</i> , <i>Prionace glauca</i>		
B		International Bottom Trawl Survey in the Mediterranean (Medits), IUCN Red List, 258,259-262
P/B	Mortality (z)= F+M	263
Q/B	$\log (Q/B) = 7.964 - 0.204 \log W - 1.965 T + 0.083 A + 0.532 h + 0.398 d$	264
D		243,265-267
B-C		67,69,70,246,268-270
14. Non-commercial large pelagic fish: <i>Cetorhinus maximus</i> , <i>Mobula mobular</i> , <i>Mola mola</i>		
B		271-274
P/B	Mortality (z)= F+M	263
Q/B	$\log (Q/B) = 7.964 - 0.204 \log W - 1.965 T + 0.083 A + 0.532 h + 0.398 d$	264
D		243,266,275
B-C		IUCN Red List, 70,246,270,276,277
15. Bluefin tuna: <i>Thunnus thynnus</i>		
B		278
P/B	Mortality (z)= F+M	263
Q/B	$\log (Q/B) = 7.964 - 0.204 \log W - 1.965 T + 0.083 A + 0.532 h + 0.398 d$	264
D		243,279-286
C		GFCM-FAO
Di		67,70,247,251,270,287,288
16. Swordfish: <i>Xiphias gladius</i>		
B		289

P/B	Mortality (z)= F+M	263
Q/B	$\log (Q/B) = 7.964 - 0.204 \log W - 1.965 T + 0.083 A + 0.532 h + 0.398 d$	264
D		243,282,290-294
C		GFCM-FAO
Di		67,70,246,247,251,287,288
17. Other large pelagic fish: <i>Coryphaena hippurus</i>, <i>Lichia amia</i>, <i>Seriola dumerili</i>, <i>Thunnus alalunga</i>, <i>Regalecus glesne</i>		
B		International Bottom Trawl Survey in the Mediterranean (Medits); 295,296,297
P/B	Mortality (z)= F+M	263
Q/B	$\log (Q/B) = 7.964 - 0.204 \log W - 1.965 T + 0.083 A + 0.532 h + 0.398 d$	264
D		243,298-305
C		GFCM-FAO
Di		70,251,287,288,306
18. Mackerel: <i>Scomber scombrus</i>, <i>Scomber colias</i>, <i>Scomberesox saurus saurus</i>, <i>Scomber spp.</i>		
B		International Bottom Trawl Survey in the Mediterranean (Medits); 307,308
P/B	Mortality (z)= F+M	263
Q/B	$\log (Q/B) = 7.964 - 0.204 \log W - 1.965 T + 0.083 A + 0.532 h + 0.398 d$	264
D		243,309-311
C		GFCM-FAO
Di		DCF (EC), 251,277,287,288,306,312-318
19. Horse mackerel: <i>Trachurus trachurus</i>, <i>Trachurus mediterraneus</i>, <i>Trachurus picturatus</i>		
B		International Bottom Trawl Survey in the Mediterranean (Medits)
P/B	Mortality (z)= F+M	263
Q/B	$\log (Q/B) = 7.964 - 0.204 \log W - 1.965 T + 0.083 A + 0.532 h + 0.398 d$	264
D		243,309,310,319-322
C		GFCM-FAO
Di		DCF (EC), 251,277,287,288,306,312-316,318,323-330
20. Other medium pelagic fish: <i>Alosa alosa</i>, <i>Alosa fallax</i>, <i>Caranx rhonchus</i>, <i>Naucrates doctor</i>, <i>Pomatomus saltatrix</i>, <i>Sarda sarda</i>, <i>Schedophilus medusophagus</i>, <i>Sphyrna sphyraena</i>		
B		International Bottom Trawl Survey in the Mediterranean (Medits)
P/B	Mortality (z)= F+M	263
Q/B	$\log (Q/B) = 7.964 - 0.204 \log W - 1.965 T + 0.083 A + 0.532 h + 0.398 d$	264
D		310,331-338
C		GFCM-FAO
Di		DCF (EC), 70,247,251,287,288,306,312-314,316,325,327,328,339,340
21. European pilchard: <i>Sardina pilchardus</i> (Adults&Juveniles)		

B		341-356
P/B	Mortality (z)= F+M	263
Q/B	$\log (Q/B) = 7.964 - 0.204 \log W - 1.965 T + 0.083 A + 0.532 h + 0.398 d$	264
D		310,357-364
C		GFCM-FAO
Di		DCF (EC), 306,312-316,318,323,324,339,365
23. European anchovy: <i>Engraulis encrasicolus</i> (Adults&Juveniles)		
B		341-343,345-356,366
P/B	Mortality (z)= F+M	263
Q/B	$\log (Q/B) = 7.964 - 0.204 \log W - 1.965 T + 0.083 A + 0.532 h + 0.398 d$	264
D		357,361,362,364,367-370
C		GFCM-FAO
Di		DCF (EC), 306,312,313-316,323,327,339,365
25. Other small pelagic fish: <i>Cubiceps gracilis</i>, <i>Dussumieria elopoides</i>, <i>Sardinella aurita</i>, <i>Sardinella maderensis</i>, <i>Syngnathus phlegon</i>, <i>Spicara flexuosa</i>, <i>Spicara maena</i>, <i>Spicara smaris</i>, <i>Sprattus sprattus</i>		
B		International Bottom Trawl Survey in the Mediterranean (Medits)
P/B	Mortality (z)= F+M	263
Q/B	$\log (Q/B) = 7.964 - 0.204 \log W - 1.965 T + 0.083 A + 0.532 h + 0.398 d$	264
D		310,319,361,362,364,371-376
C		GFCM-FAO
Di		DCF (EC), 277,306,312,313,315,317,318,323,324,326-330,340,377,378
26. Benthopelagic fish: <i>Argyrosomus regius</i>, <i>Coelorinchus caelorhincus</i>, <i>Dysomma brevirostre</i>, <i>Eretmophorus kleinenbergi</i>, <i>Gadella maraldi</i>, <i>Gaidropsarus biscayensis</i>, <i>Hoplostethus mediterraneus</i>, <i>Hymenocephalus italicus</i>, <i>Lagocephalus lagocephalus</i>, <i>Lagocephalus scleratus</i>, <i>Lagocephalus suezensis</i>, <i>Lepidopus caudatus</i>, <i>Nezumia aequalis</i>, <i>Nezumia sclerorhynchus</i>, <i>Physiculus dalwigki</i>, <i>Schedophilus ovalis</i>, <i>Sphoeroides pachygaster</i>, <i>Stromateus fiatola</i>, <i>Trichiurus lepturus</i>		
B		International Bottom Trawl Survey in the Mediterranean (Medits)
P/B	Mortality (z)= F+M	263
Q/B	$\log (Q/B) = 7.964 - 0.204 \log W - 1.965 T + 0.083 A + 0.532 h + 0.398 d$	264
D		243,310,379-386
C		GFCM-FAO
Di		306,312,323,324,329,340,377,387-390
27. No commercial meso (bathy) pelagic fish: <i>Arctozenus risso</i>, <i>Argyrolepecus aculeatus</i>, <i>Argyrolepecus hemigymnus</i>, <i>Bathophilus nigerrimus</i>, <i>Benthosema glaciale</i>, <i>Brama brama</i>, <i>Centrolophus niger</i>, <i>Ceratoscopelus maderensis</i>, <i>Chauliodus sloani</i>, <i>Cyclothone braueri</i>, <i>Cyclothone pygmaea</i>, <i>Diaphus holti</i>, <i>Diaphus metopoclampus</i>, <i>Diaphus rafinesquii</i>, <i>Electrona risso</i>, <i>Evermannella balbo</i>, <i>Gadiculus argenteus</i>, <i>Gonichthys cocco</i>, <i>Gonostoma denudatum</i>, <i>Hygophum benoitii</i>, <i>Hygophum hygomi</i>, <i>Ichthyococcus ovatus</i>, <i>Lampanyctus crocodilus</i>, <i>Lampanyctus pusillus</i>, <i>Lepidion lepidion</i>, <i>Lestidiops sphyrenoides</i>, <i>Lobianchia dofleini</i>, <i>Lobianchia gemellarii</i>, <i>Maurollicus muelleri</i>, <i>Melanostigma atlanticum</i>, <i>Mora moro</i>, <i>Myctophidae</i>, <i>Myctophum punctatum</i>, <i>Nansenia oblita</i>, <i>Nemichthys scolopaceus</i>, <i>Nettastoma melanurum</i>, <i>Notacanthus bonaparte</i>, <i>Notoscopelus bolini</i>, <i>Notoscopelus elongatus</i>, <i>Paralepis coregonoides</i>, <i>Psenes pellucidus</i>, <i>Rhynchogadus hepaticus</i>, <i>Stomias boa boa</i>, <i>Sudis hyalina</i>, <i>Symbolophorus veranyi</i>, <i>Trachipterus trachipterus</i>, <i>Vinciguerrria attenuata</i>, <i>Vinciguerrria poweriae</i>		

B		International Bottom Trawl Survey in the Mediterranean (Medit)
P/B	Mortality (z)= F+M	263
Q/B	$\log (Q/B) = 7.964 - 0.204 \log W - 1.965 T + 0.083 A + 0.532 h + 0.398 d$	264
D		382,391-400
C		GFCM-FAO
Di		70,312,323,324,329,340,377,389,401
28. Anglerfish: <i>Lophius budegassa</i>, <i>Lophius piscatorius</i>		
B		International Bottom Trawl Survey in the Mediterranean (Medit)
P/B	Mortality (z)= F+M	263
Q/B	$\log (Q/B) = 7.964 - 0.204 \log W - 1.965 T + 0.083 A + 0.532 h + 0.398 d$	264
D		310,382,402-406
C		GFCM-FAO
Di		DCF (EC), 277,306,313-315,317,323,324,328-330,401
29. European hake: <i>Merluccius merluccius</i> (Adults&Juveniles)		
B		International Bottom Trawl Survey in the Mediterranean (Medit)
P/B	Mortality (z)= F+M	263
Q/B	$\log (Q/B) = 7.964 - 0.204 \log W - 1.965 T + 0.083 A + 0.532 h + 0.398 d$	264
D		310,382,407-414
C		GFCM-FAO
Di		DCF (EC), 277,306,313-315,323,324,328-330
31. Other commercial large demersal fish: <i>Conger conger</i>, <i>Epinephelus aeneus</i>, <i>Epinephelus costae</i>, <i>Epinephelus marginatus</i>, <i>Molva dypterygia</i>, <i>Molva macrophthalma</i>, <i>Molva molva</i>, <i>Polyprion americanus</i>		
B		International Bottom Trawl Survey in the Mediterranean (Medit)
P/B	Mortality (z)= F+M	263
Q/B	$\log (Q/B) = 7.964 - 0.204 \log W - 1.965 T + 0.083 A + 0.532 h + 0.398 d$	264
D		310,382,415-423
C		GFCM-FAO
Di		DCF (EC), 277,306,312-314,323,324,329,330,389,401,424
32. Gadidae: <i>Merlangius merlangus</i>, <i>Micromesistius poutassou</i>, <i>Trisopterus capelanus</i>, <i>Trisopterus luscus</i>		
B		International Bottom Trawl Survey in the Mediterranean (Medit)
P/B	Mortality (z)= F+M	263
Q/B	$\log (Q/B) = 7.964 - 0.204 \log W - 1.965 T + 0.083 A + 0.532 h + 0.398 d$	264
D		310,382,425-433

C		GFCM-FAO
Di		DCF (EC), 277,312-315,323,324,327-330,389,401
33. Sparidae: <i>Boops boops</i> , <i>Centracanthus cirrus</i> , <i>Dentex dentex</i> , <i>Dentex gibbosus</i> , <i>Dentex macrophthalmus</i> , <i>Dentex maroccanus</i> , <i>Diplodus annularis</i> , <i>Diplodus cervinus</i> , <i>Diplodus puntazzo</i> , <i>Diplodus sargus</i> , <i>Diplodus vulgaris</i> , <i>Lithognathus mormyrus</i> , <i>Oblada melanura</i> , <i>Pagellus acarne</i> , <i>Pagellus bogaraveo</i> , <i>Pagellus erythrinus</i> , <i>Pagrus caeruleostictus</i> , <i>Pagrus pagrus</i> , <i>Sarpa salpa</i> , <i>Sparus aurata</i> , <i>Spondylus cantharus</i>		
B		International Bottom Trawl Survey in the Mediterranean (Medit)
P/B	Mortality (z)= F+M	263
Q/B	$\log (Q/B) = 7.964 - 0.204 \log W - 1.965 T + 0.083 A + 0.532 h + 0.398 d$	264
D		310,319,414,419,428,434-463
C		GFCM-FAO
Di		DCF (EC), 277,306,312-318,323,324,326,329,330,339,401
34. Other commercial medium demersal fish: <i>Anguilla anguilla</i> , <i>Chelidonichthys cuculus</i> , <i>Chelidonichthys lucerna</i> , <i>Chelidonichthys obscurus</i> , <i>Chelon labrosus</i> , <i>Dactylopterus volitans</i> , <i>Dicentrarchus labrax</i> , <i>Eutrigla gurnardus</i> , <i>Gaidropsarus mediterraneus</i> , <i>Gaidropsarus vulgaris</i> , <i>Gnathophis mystax</i> , <i>Helicolenus dactylopterus</i> , <i>Labrus mixtus</i> , <i>Labrus viridis</i> , <i>Liza aurata</i> , <i>Liza ramada</i> , <i>Liza saliens</i> , <i>Mugil cephalus</i> , <i>Oedalechilus labeo</i> , <i>Parapristipoma octolineatum</i> , <i>Peristedion cataphractum</i> , <i>Phycis blennoides</i> , <i>Phycis phycis</i> , <i>Sciaena umbra</i> , <i>Scorpaena elongata</i> , <i>Scorpaena loppei</i> , <i>Scorpaena maderensis</i> , <i>Scorpaena notata</i> , <i>Scorpaena porcus</i> , <i>Scorpaena scrofa</i> , <i>Serranus atricauda</i> , <i>Serranus cabrilla</i> , <i>Serranus hepatus</i> , <i>Serranus scriba</i> , <i>Sparisoma cretense</i> , <i>Symphodus tinca</i> , <i>Trachinus radiatus</i> , <i>Trachyscorpia cristulata echinata</i> , <i>Trigla lyra</i> , <i>Trigloporus lastoviza</i> , <i>Umbrina canariensis</i> , <i>Umbrina cirrosa</i> , <i>Umbrina ronchus</i> , <i>Xyrichtys novacula</i> , <i>Zeus faber</i>		
B		International Bottom Trawl Survey in the Mediterranean (Medit)
P/B	Mortality (z)= F+M	263
Q/B	$\log (Q/B) = 7.964 - 0.204 \log W - 1.965 T + 0.083 A + 0.532 h + 0.398 d$	264
D		310,319,382,419,438,443,448,461,464-495
C		GFCM-FAO
Di		DCF (EC), 247,277,306,312,313,315,317,318,323,324,326-330,377,389,390,401,496
35. Flatfish: <i>Arnoglossus imperialis</i> , <i>Arnoglossus kessleri</i> , <i>Arnoglossus laterna</i> , <i>Arnoglossus rueppelii</i> , <i>Arnoglossus thori</i> , <i>Bathysolea profundicola</i> , <i>Bothus podas</i> , <i>Buglossidium luteum</i> , <i>Citharus linguatula</i> , <i>Dicologlossa cuneata</i> , <i>Dicologlossa hexophthalma</i> , <i>Lepidorhombus boscii</i> , <i>Lepidorhombus whiffiagonis</i> , <i>Microchirus azevia</i> , <i>Microchirus boscanion</i> , <i>Microchirus ocellatus</i> , <i>Microchirus variegatus</i> , <i>Monochirus hispidus</i> , <i>Platichthys flesus</i> , <i>Pegusa impar</i> , <i>Pegusa lascaris</i> , <i>Scophthalmus maximus</i> , <i>Scophthalmus rhombus</i> , <i>Solea senegalensis</i> , <i>Solea solea</i> , <i>Solea spp.</i> , <i>Symphurus ligulatus</i> , <i>Symphurus nigrescens</i> , <i>Synapturichthys kleinii</i> , <i>Zeugopterus regius</i>		
B		International Bottom Trawl Survey in the Mediterranean (Medit)
P/B	Mortality (z)= F+M	263
Q/B	$\log (Q/B) = 7.964 - 0.204 \log W - 1.965 T + 0.083 A + 0.532 h + 0.398 d$	264
D		310,319,382,406,419,431,448,497-508
C		GFCM-FAO
Di		DCF (EC), 247,277,306,312,313,323,324,326,327,329,330,339,377,389,401,496
36. No commercial medium demersal fish: <i>Apterichthys caecus</i> , <i>Aulopus filamentosus</i> , <i>Balistes capriscus</i> , <i>Cepola macrophthalma</i> , <i>Chlopsis bicolor</i> , <i>Dalophis imberbis</i> , <i>Echelus myrus</i> , <i>Facciolella oxyrhyncha</i> , <i>Fistularia commersonii</i> , <i>Muraena helena</i> , <i>Nerophis maculatus</i> , <i>Ophichthys rufus</i> , <i>Ophidion barbatum</i> , <i>Ophidion rochei</i> , <i>Ophisurus serpens</i> , <i>Saurida undosquamis</i> , <i>Siganus luridus</i> , <i>Siganus rivulatus</i> , <i>Stephanolepis diaspros</i> , <i>Syngnathus acus</i> , <i>Syngnathus spp.</i> , <i>Syngnathus tenuirostris</i> , <i>Syngnathus typhle</i> , <i>Synodus saurus</i> , <i>Trachinus araneus</i> , <i>Trachinus draco</i> , <i>Uranoscopus scaber</i>		

B		International Bottom Trawl Survey in the Mediterranean (Medit)
P/B	Mortality (z)= F+M	263
Q/B	$\log (Q/B) = 7.964 - 0.204 \log W - 1.965 T + 0.083 A + 0.532 h + 0.398 d$	264
D		310,319,419,448,474,487,509-518
Di		247,277,306,313,324,326,327,330,377,401,496
37. Red mullet: <i>Mullus barbatus</i> (Adults&Juveniles)		
B		International Bottom Trawl Survey in the Mediterranean (Medit)
P/B	Mortality (z)= F+M	263
Q/B	$\log (Q/B) = 7.964 - 0.204 \log W - 1.965 T + 0.083 A + 0.532 h + 0.398 d$	264
D		310,448,462,509,519-525
C		GFCM-FAO
Di		DCF (EC), 306,312,313-315,317,326,330,401
39. Surmullet: <i>Mullus surmuletus</i>		
B		International Bottom Trawl Survey in the Mediterranean (Medit)
P/B	Mortality (z)= F+M	263
Q/B	$\log (Q/B) = 7.964 - 0.204 \log W - 1.965 T + 0.083 A + 0.532 h + 0.398 d$	264
D		310,319,419,448,462,509,518,519,523,526,527
C		GFCM-FAO
Di		DCF (EC), 277,306,312,313,315,317,326,377,401
40. Commercial small demersal fish: <i>Acantholabrus palloni</i>, <i>Anthias anthias</i>, <i>Aphia minuta</i>, <i>Ariosoma balearicum</i>, <i>Atherina boyeri</i>, <i>Coris julis</i>, <i>Gymnamodytes cicereus</i>, <i>Lappanella fasciata</i>, <i>Lepidotrigla cavillone</i>, <i>Lepidotrigla dieuzeidei</i>, <i>Symphodus cinereus</i>, <i>Symphodus doderleini</i>, <i>Symphodus mediterraneus</i>, <i>Symphodus ocellatus</i>, <i>Symphodus roissali</i>, <i>Symphodus rostratus</i>		
B		International Bottom Trawl Survey in the Mediterranean (Medit)
P/B	Mortality (z)= F+M	263
Q/B	$\log (Q/B) = 7.964 - 0.204 \log W - 1.965 T + 0.083 A + 0.532 h + 0.398 d$	264
D		310,319,419,448,467,474,528-533
C		GFCM-FAO
Di		DCF (EC), 247,277,306,312-314,317,318,323,327,329,534
41. No commercial small demersal fish: <i>Apogon imberbis</i>, <i>Blenniidae</i>, <i>Blennius ocellaris</i>, <i>Callanthias ruber</i>, <i>Callionymus lyra</i>, <i>Callionymus pusillus</i>, <i>Callionymus spp.</i>, <i>Capros aper</i>, <i>Carapus acus</i>, <i>Chlorophthalmus agassizi</i>, <i>Chromis chromis</i>, <i>Coryphoblennius galerita</i>, <i>Crystalllogobius linearis</i>, <i>Deltentosteus collonianus</i>, <i>Deltentosteus quadrimaculatus</i>, <i>Diplecogaster bimaculata</i>, <i>Echiodon dentatus</i>, <i>Gobius cobitis</i>, <i>Gobius cruentatus</i>, <i>Gobius fallax</i>, <i>Gobius geniporus</i>, <i>Gobius niger</i>, <i>Gobius paganellus</i>, <i>Gobius spp.</i>, <i>Grammonus ater</i>, <i>Hippocampus guttulatus</i>, <i>Hippocampus hippocampus</i>, <i>Hippocampus ramulosus</i>, <i>Lepadogaster lepadogaster</i>, <i>Lepadogaster spp.</i>, <i>Lesueurigobius friesii</i>, <i>Lesueurigobius sanzi</i>, <i>Lesueurigobius suerii</i>, <i>Macroramphosus scolopax</i>, <i>Microlipophrys adriaticus</i>, <i>Microichthys coccoi</i>, <i>Parablennius gattorugine</i>, <i>Parablennius incognitus</i>, <i>Parablennius tentacularis</i>, <i>Pomatoschistus marmoratus</i>, <i>Pomatoschistus microps</i>, <i>Pomatoschistus minutus</i>, <i>Pomatoschistus norvegicus</i>, <i>Pomatoschistus spp.</i>, <i>Pteragogus pelycus</i>, <i>Salaria pavo</i>, <i>Synchiropus phaeton</i>, <i>Syngnathus abaster</i>, <i>Syngnathus taenionotus</i>, <i>Upeneus moluccensis</i>, <i>Upeneus pori</i>		

B		International Bottom Trawl Survey in the Mediterranean (Medit)
P/B	Mortality (z)= F+M	263
Q/B	$\log (Q/B) = 7.964 - 0.204 \log W - 1.965 T + 0.083 A + 0.532 h + 0.398 d$	264
D		310,382,419,448,462,474,509,535-542
Di		247,306,312,313,318,324,326,327,329,390,534
42. Bathydemersal (deep sea) fish: <i>Alepocephalus rostratus</i> , <i>Argentina sphyraena</i> , <i>Bathypterois dubius</i> , <i>Bellottia apoda</i> , <i>Benthocometes robustus</i> , <i>Borostomias antarcticus</i> , <i>Cataetx alleni</i> , <i>Coelorinchus occa</i> , <i>Epigonus constanciae</i> , <i>Epigonus denticulatus</i> , <i>Epigonus telescopus</i> , <i>Eutelichthys leptochirus</i> , <i>Glossanodon leioglossus</i> , <i>Paraliparis murieli</i> , <i>Polyacanthonotus rissoanus</i> , <i>Trachyrincus scabrus</i>		
B		International Bottom Trawl Survey in the Mediterranean (Medit)
P/B	Mortality (z)= F+M	263
Q/B	$\log (Q/B) = 7.964 - 0.204 \log W - 1.965 T + 0.083 A + 0.532 h + 0.398 d$	264
D		317,382,394,395,536,543-546
C		GFCM-FAO
Di		312,323,324,329,389
43. Small-spotted catshark: <i>Scyliorhinus canicula</i>		
B		International Bottom Trawl Survey in the Mediterranean (Medit)
P/B	Mortality (z)= F+M	263
Q/B	$\log (Q/B) = 7.964 - 0.204 \log W - 1.965 T + 0.083 A + 0.532 h + 0.398 d$	264
D		310,382,448,547-552
C		GFCM-FAO
Di		DCF (EC), 247,277,312,315,317,323,324,329,330,377,389,401
44. Blackmouth catshark: <i>Galeus melastomus</i>		
B		International Bottom Trawl Survey in the Mediterranean (Medit)
P/B	Mortality (z)= F+M	263
Q/B	$\log (Q/B) = 7.964 - 0.204 \log W - 1.965 T + 0.083 A + 0.532 h + 0.398 d$	264
D		310,382,551-554
C		GFCM-FAO
Di		DCF (EC), 312,315,317,323,324,329,377,389
45. Other small demersal sharks: <i>Centrophorus granulosus</i> , <i>Chimaera monstrosa</i> , <i>Dalatias licha</i> , <i>Etmopterus spinax</i> , <i>Galeus atlanticus</i> , <i>Heptranchias perlo</i> , <i>Hexanchus griseus</i> , <i>Hiatella arctica</i> , <i>Mustelus asterias</i> , <i>Mustelus mustelus</i> , <i>Mustelus punctulatus</i> , <i>Oxynotus centrina</i> , <i>Scyliorhinus stellaris</i> , <i>Squalus acanthias</i> , <i>Squalus blainville</i> , <i>Squalus uyato</i> , <i>Squatina aculeata</i> , <i>Squatina oculata</i> , <i>Squatina squatina</i>		
B		International Bottom Trawl Survey in the Mediterranean (Medit)
P/B	Mortality (z)= F+M	263
Q/B	$\log (Q/B) = 7.964 - 0.204 \log W - 1.965 T + 0.083 A + 0.532 h + 0.398 d$	264

D		266,310,382,448,549,551,553,555-566
C		GFCM-FAO
Di		DCF (EC), 277,306,312,313,315,317,323,324,329,377,389,424
46. Rays and skates: <i>Dasyatis centroura</i> , <i>Dasyatis pastinaca</i> , <i>Dasyatis tortonesei</i> , <i>Dipturus batis</i> , <i>Dipturus nidarosiensis</i> , <i>Dipturus oxyrinchus</i> , <i>Gymnura altavela</i> , <i>Leucoraja circularis</i> , <i>Leucoraja fullonica</i> , <i>Leucoraja melitensis</i> , <i>Leucoraja naevus</i> , <i>Myliobatis aquila</i> , <i>Pteromylaeus bovinus</i> , <i>Pteroplatytrygon violacea</i> , <i>Raja asterias</i> , <i>Raja brachyura</i> , <i>Raja clavata</i> , <i>Raja miraletus</i> , <i>Raja montagui</i> , <i>Raja polystigma</i> , <i>Raja radula</i> , <i>Raja spp.</i> , <i>Raja undulata</i> , <i>Rostroraja alba</i>		
B		International Bottom Trawl Survey in the Mediterranean (Medit)
P/B	Mortality (z)= F+M	263
Q/B	$\log (Q/B) = 7.964 - 0.204 \log W - 1.965 T + 0.083 A + 0.532 h + 0.398 d$	264
D		266,310,406,448,549,551,552,567-579
C		GFCM-FAO
Di		DCF (EC), 70,247,277,306,312,313,315,317,323,324,329,330,339,377,401,424
47. Torpedos: <i>Torpedo marmorata</i> , <i>Torpedo nobiliana</i> , <i>Torpedo spp.</i> , <i>Torpedo torpedo</i>		
B		International Bottom Trawl Survey in the Mediterranean (Medit)
P/B	Mortality (z)= F+M	263
Q/B	$\log (Q/B) = 7.964 - 0.204 \log W - 1.965 T + 0.083 A + 0.532 h + 0.398 d$	264
D		266,310,448,518,580,581
C		GFCM-FAO
Di		247,277,306,314,315,317,324,329,330,401,496
48. Benthic cephalopods: <i>Bathypolypus sponsalis</i> , <i>Callistoctopus macropus</i> , <i>Eledone cirrhosa</i> , <i>Eledone moschata</i> , <i>Macrotritopus defilippi</i> , <i>Neorossia caroli</i> , <i>Octopus salutii</i> , <i>Octopus vulgaris</i> , <i>Pteroctopus tetracirrhus</i> , <i>Rondeletiola minor</i> , <i>Rossia macrosoma</i> , <i>Rossia spp.</i> , <i>Scaevurgus unicolor</i> , <i>Sepia elegans</i> , <i>Sepia officinalis</i> , <i>Sepia orbignyana</i> , <i>Sepietta neglecta</i> , <i>Sepietta obscura</i> , <i>Sepietta oweniana</i> , <i>Sepietta spp.</i> , <i>Sepioloa affinis</i> , <i>Sepioloa intermedia</i> , <i>Sepioloa ligulata</i> , <i>Sepioloa robusta</i> , <i>Sepioloa rondeletii</i> , <i>Sepioloa spp.</i>		
B		International Bottom Trawl Survey in the Mediterranean (Medit)
P/B	$\log M = -0.2107 - 0.0824 \log W + 0.6757 \log K + 0.4627 \log T$	582
Q/B		583
D		552,584-593
C		GFCM-FAO
Di		DCF (EC), 277,306,312,313,317,324,326,329,377,389,401,534,552
49. Benthopelagic cephalopods: <i>Abralia veranyi</i> , <i>Abraliopsis morisii</i> , <i>Alloteuthis media</i> , <i>Alloteuthis spp.</i> , <i>Alloteuthis subulata</i> , <i>Illex coindetii</i> , <i>Loligo forbesii</i> , <i>Loligo spp.</i> , <i>Loligo vulgaris</i> , <i>Ocythoe tuberculata</i> , <i>Opisthoteuthis calypso</i> , <i>Todarodes sagittatus</i> , <i>Todaropsis eblanae</i>		
B		International Bottom Trawl Survey in the Mediterranean (Medit)
P/B	$\log M = -0.2107 - 0.0824 \log W + 0.6757 \log K + 0.4627 \log T$	582
Q/B		583

D		552,594-601
C		GFCM-FAO
Di		DCF (EC), 277,306,312,313,315,317,318,324,327,329,377,389,401,496
50. Mesopelagic cephalopods: <i>Ancistrocheirus lesueurii</i> , <i>Ancistroteuthis lichtensteini</i> , <i>Argonauta argo</i> , <i>Brachioeuthis riisei</i> , <i>Chiroteuthis veranii</i> , <i>Ctenopteryx sicula</i> , <i>Heteroteuthis dispar</i> , <i>Histioteuthis bonnellii</i> , <i>Histioteuthis reversa</i> , <i>Histioteuthis spp.</i> , <i>Octopoteuthis sicula</i> , <i>Onychoteuthis banksii</i> , <i>Pyroteuthis margaritifera</i> , <i>Stoloteuthis leucoptera</i>		
B		International Bottom Trawl Survey in the Mediterranean (Medit)
P/B	$\log M = -0.2107 - 0.0824 \log W + 0.6757 \log K + 0.4627 \log T$	582
Q/B		583
D		552,598,602
Di		70,312,324,329,389
51. Bivalves: <i>Abra alba</i> , <i>Abra longicallus</i> , <i>Acanthocardia aculeata</i> , <i>Acanthocardia echinata</i> , <i>Acanthocardia paucicostata</i> , <i>Acanthocardia spinosa</i> , <i>Acanthocardia tuberculata</i> , <i>Aequipecten commutatus</i> , <i>Aequipecten opercularis</i> , <i>Amygdalum politum</i> , <i>Anadara corbuloides</i> , <i>Anadara inaequivalvis</i> , <i>Anadara transversa</i> , <i>Anomia ephippium</i> , <i>Arca noae</i> , <i>Arca tetragona</i> , <i>Arcopagia balaustina</i> , <i>Arcopagia crassa</i> , <i>Atrina fragilis</i> , <i>Atrina pectinata</i> , <i>Callista chione</i> , <i>Cardiomya spp.</i> , <i>Chama gryphoides</i> , <i>Chamelea gallina</i> , <i>Clausinella fasciata</i> , <i>CrassOweniidae</i> <i>gigas</i> , <i>Cuspidaria cuspidata</i> , <i>Flexopecten flexuosus</i> , <i>Flexopecten glaber glaber</i> , <i>Gibbomodiola adriatica</i> , <i>Glossus humanus</i> , <i>Gryphus vitreus</i> , <i>Hiatella arctica</i> , <i>Laevicardium crassum</i> , <i>Laevicardium oblongum</i> , <i>Lima lima</i> , <i>Limaria hians</i> , <i>Lutraria spp.</i> , <i>Mimachlamys varia</i> , <i>Modiolus modiolus</i> , <i>Monia squama</i> , <i>Musculus subpictus</i> , <i>Mytilus edulis</i> , <i>Mytilus galloprovincialis</i> , <i>Neopercnodonte cochlear</i> , <i>Nucula nucleus</i> , <i>Nucula sulcata</i> , <i>Ostrea edulis</i> , <i>Pecten jacobaeus</i> , <i>Pecten maximus</i> , <i>Pinna nobilis</i> , <i>Pinna rudis</i> , <i>Pseudamussium clavatum</i> , <i>Pteria hirundo</i> , <i>Ruditapes decussatus</i> , <i>Solecurtus scopula</i> , <i>Spisula subtruncata</i> , <i>Tellina fabula</i> , <i>Tellina planata</i> , <i>Teredo navalis</i> , <i>Thracia phaseolina</i> , <i>Venus casina</i> , <i>Venus nux</i> , <i>Venus verrucosa</i>		
B		International Bottom Trawl Survey in the Mediterranean (Medit)
P/B	$\log(P/B) = 10.154 - 0.271 * \text{LOG}(W) - 2824.247 * (1/(BT+273)) - 0.063 * \text{LOG}(D+1) + 0.13 * (Me) + 0.076 * (Car) - 0.311 * (Mol) - 0.154 * (Crus) - 0.266 * (Pol) - 0.472 * (Echi)$	603,604
Q/B	(P/B)/(P/Q)	
D		605-608
C		GFCM-FAO
Di		312,313,324,329,339,377,534
52. Gastropods: <i>Alcyonidium spp.</i> , <i>Aplysia depilans</i> , <i>Aplysia fasciata</i> , <i>Aplysia punctata</i> , <i>Aplysia parvula</i> , <i>Aporrhais pespelecani</i> , <i>Aporrhais serresianus</i> , <i>Armina loveni</i> , <i>Armina maculata</i> , <i>Armina neapolitana</i> , <i>Armina tigrina</i> , <i>Berthella aurantiaca</i> , <i>Bivetiella cancellata</i> , <i>Bolinus brandaris</i> , <i>Bolma rugosa</i> , <i>Buccinum humphreysianum</i> , <i>Calliostoma granulatum</i> , <i>Calliostoma laugierii</i> , <i>Calliostoma zizyphinum</i> , <i>Callumbonella suturalis</i> , <i>Calyptraea chinensis</i> , <i>Capulus ungaricus</i> , <i>Cavolinia tridentata</i> , <i>Coralliophila meyendorffii</i> , <i>Crepidula unguiformis</i> , <i>Charonia lampas</i> , <i>Colus gracilis</i> , <i>Colus jeffreysianus</i> , <i>Cylichna cylindracea</i> , <i>Cymbium olla</i> , <i>Cymbulia peronii</i> , <i>Dendrodoris limbata</i> , <i>Diodora italica</i> , <i>Doris pseudoargus</i> , <i>Doris sticta</i> , <i>Doris verrucosa</i> , <i>Epitonium clathratulum</i> , <i>Epitonium clathrus</i> , <i>Euspira catena</i> , <i>Euspira fusca</i> , <i>Euspira grossularia</i> , <i>Euspira guilleminii</i> , <i>Euthria cornea</i> , <i>Felimare picta</i> , <i>Fusinus rostratus</i> , <i>Fusinus syracusanus</i> , <i>Fusiturrus similis</i> , <i>Fusiturrus undatiruga</i> , <i>Galeodea echinophora</i> , <i>Galeodea rugosa</i> , <i>Gastroperon rubrum</i> , <i>Gibbula magus</i> , <i>Hadriania craticulata</i> , <i>Haminoea hydatis</i> , <i>Haminoea navicula</i> , <i>Hexaplex trunculus</i> , <i>Hirtomurex squamosus</i> , <i>Jorunna tomentosa</i> , <i>Kaloplocamus ramosus</i> , <i>Lamellaria perspicua</i> , <i>Marionia blainvillea</i> , <i>Monoplex corrugatus</i> , <i>Nassarius incrassatus</i> , <i>Nassarius lima</i> , <i>Nassarius reticulatus</i> , <i>Naticarius hebraeus</i> , <i>Naticarius stercusmuscarum</i> , <i>Ocenebra erinaceus</i> , <i>Okenia elegans</i> , <i>Peliodoris atromaculata</i> , <i>Philine aperta</i> , <i>Philineopsis depicta</i> , <i>Platydoris argo</i> , <i>Pleurobranchaea meckeli</i> , <i>Pleurobranchus membranaceus</i> , <i>Pleurobranchus testudinarius</i> , <i>Pseudosimnia adriatica</i> , <i>Pseudosimnia carnea</i> , <i>Ranella olearium</i> , <i>Scaphander lignarius</i> , <i>Semicassis granulata</i> , <i>Semicassis saburon</i> , <i>Tenagodus obtusus</i> , <i>Tethys fimbria</i> , <i>Tonna galea</i> , <i>Tritonia hombergii</i> , <i>Trophonopsis muricata</i> , <i>Turritella communis</i> , <i>Turritella turbona</i> , <i>Umbraculum umbraculum</i> , <i>Xenophora crispa</i>		
B		International Bottom Trawl Survey in the Mediterranean (Medit)

P/B	$\log(P/B) = 10.154 - 0.271 * \log(M) - 2824.247 * (1/(T+273)) - 0.063 * \log(D+1) + 0.13 * (Me) + 0.076 * (Car) - 0.311 * (Mol) - 0.154 * (Crus) - 0.266 * (Pol) - 0.472 * (Echi)$	603,604
Q/B	(P/B)/(P/Q)	
D		609-613
C		GFCM-FAO
Di		312-314,324,329,339,534
53. Deep-water rose shrimp: <i>Parapenaeus longirostris</i>		
B		International Bottom Trawl Survey in the Mediterranean (Medit)
P/B	$\log(P/B) = 10.154 - 0.271 * \log(M) - 2824.247 * (1/(T+273)) - 0.063 * \log(D+1) + 0.13 * (Me) + 0.076 * (Car) - 0.311 * (Mol) - 0.154 * (Crus) - 0.266 * (Pol) - 0.472 * (Echi)$	603,604
Q/B		583
D		614-617
C		GFCM-FAO
Di		DCF (EC), 306,313,317,323,324,329,377,378,389,534
54. Blue and red shrimp: <i>Aristeus antennatus</i>		
B		International Bottom Trawl Survey in the Mediterranean (Medit)
P/B	$\log(P/B) = 10.154 - 0.271 * \log(M) - 2824.247 * (1/(T+273)) - 0.063 * \log(D+1) + 0.13 * (Me) + 0.076 * (Car) - 0.311 * (Mol) - 0.154 * (Crus) - 0.266 * (Pol) - 0.472 * (Echi)$	603,604
Q/B		583
D		616,618-622
C		GFCM-FAO
Di		DCF (EC), 306,315,317,377,534
55. Giant red shrimp: <i>Aristaeomorpha foliacea</i>		
B		International Bottom Trawl Survey in the Mediterranean (Medit)
P/B	$\log(P/B) = 10.154 - 0.271 * \log(M) - 2824.247 * (1/(T+273)) - 0.063 * \log(D+1) + 0.13 * (Me) + 0.076 * (Car) - 0.311 * (Mol) - 0.154 * (Crus) - 0.266 * (Pol) - 0.472 * (Echi)$	603,604
Q/B		583
D		614,622-628
C		GFCM-FAO
Di		DCF (EC), 306,317,323,324,329,377,378,534

56. Other commercial shrimps: <i>Penaeus kerathurus</i> , <i>Penaeus japonicus</i>		
B		International Bottom Trawl Survey in the Mediterranean (Medit)
P/B	$\log(P/B) = 10.154 - 0.271 * \log(M) - 2824.247 * (1/(T+273)) - 0.063 * \log(D+1) + 0.13 * (Me) + 0.076 * (Car) - 0.311 * (Mol) - 0.154 * (Crus) - 0.266 * (Pol) - 0.472 * (Echi)$	603,604
Q/B		583
D		629
C		GFCM-FAO
Di		DCF (EC), 313,377,390,534
57. No commercial shrimps: <i>AcanthePHYra eximia</i> , <i>AcanthePHYra pelagica</i> , <i>Aegaeon cataphractus</i> , <i>Aegaeon lacazei</i> , <i>Allosergestes sargassi</i> , <i>Alpheus glaber</i> , <i>Alpheus macrocheles</i> , <i>Alpheus platydactylus</i> , <i>Ascidonia flavomaculata</i> , <i>Athanas nitescens</i> , <i>Calocarides coronatus</i> , <i>Calocaris macandreae</i> , <i>Caridion steveni</i> , <i>Chlorotocus crassicornis</i> , <i>Crangon</i> , <i>Eusergestes arcticus</i> , <i>Funchalia woodwardi</i> , <i>Gennadas elegans</i> , <i>Hymenopenaeus debilis</i> , <i>Ligur ensiferus</i> , <i>Lysmata seticaudata</i> , <i>Lophogaster typicus</i> , <i>Pandalina brevirostris</i> , <i>Pandalina profunda</i> , <i>Pasiphaea multidentata</i> , <i>Pasiphaea sivado</i> , <i>Penaeopsis serrata</i> , <i>Periclimenes granulatus</i> , <i>Philocheras echinulatus</i> , <i>Philocheras sculptus</i> , <i>Plesionika acanthonotus</i> , <i>Plesionika antigai</i> , <i>Plesionika edwardsii</i> , <i>Plesionika giglioli</i> , <i>Plesionika heterocarpus</i> , <i>Plesionika martia</i> , <i>Plesionika narval</i> , <i>Pontophilus norvegicus</i> , <i>Pontophilus spinosus</i> , <i>Processa canaliculata</i> , <i>Processa edulis edulis</i> , <i>Processa elegantula</i> , <i>Processa noveli</i> , <i>Richardina fredericii</i> , <i>Sergestes arachnipedus</i> , <i>Sergia robusta</i> , <i>Sicyonia carinata</i> , <i>Solenocera membranacea</i> , <i>Systellaspis debilis</i> , <i>Typton spongicola</i> , <i>Stenopus spinosus</i> , <i>Synalpheus gambarelloides</i> , <i>Upogebia pusilla</i>		
B		International Bottom Trawl Survey in the Mediterranean (Medit)
P/B	$\log(P/B) = 10.154 - 0.271 * \log(M) - 2824.247 * (1/(T+273)) - 0.063 * \log(D+1) + 0.13 * (Me) + 0.076 * (Car) - 0.311 * (Mol) - 0.154 * (Crus) - 0.266 * (Pol) - 0.472 * (Echi)$	603,604
Q/B		583
D		614,630-639
C		GFCM-FAO
Di		313,324,329,378,389,534
58. Norway lobster: <i>Nephrops norvegicus</i>		
B		International Bottom Trawl Survey in the Mediterranean (Medit)
P/B	$\log(P/B) = 10.154 - 0.271 * \log(M) - 2824.247 * (1/(T+273)) - 0.063 * \log(D+1) + 0.13 * (Me) + 0.076 * (Car) - 0.311 * (Mol) - 0.154 * (Crus) - 0.266 * (Pol) - 0.472 * (Echi)$	603,604
Q/B		583
D		640-642
C		GFCM-FAO
Di		DCF (EC), 313,317,324,329,377,378,389,390
59. Other commercial decapods: <i>Homarus gammarus</i> , <i>Maja squinado</i> , <i>Necora puber</i> , <i>Palinurus elephas</i> , <i>Palinurus mauritanicus</i> , <i>Palinurus spp</i> , <i>Squilla mantis</i>		
B		International Bottom Trawl Survey in the Mediterranean (Medit)

P/B	$\log(P/B) = 10.154 - 0.271 * \log(M) - 2824.247 * (1/(T+273)) - 0.063 * \log(D+1) + 0.13 * (Me) + 0.076 * (Car) - 0.311 * (Mol) - 0.154 * (Crus) - 0.266 * (Pol) - 0.472 * (Echi)$	603,604
Q/B		583
D		643-646
C		GFCM-FAO
Di		DCF (EC), 247,306,313,326,329,339,401,534
60. Non-commercial decapods: <i>Anamathia rissoana</i> , <i>Anapagurus bicorniger</i> , <i>Anapagurus laevis</i> , <i>Atelecyclotus rotundatus</i> , <i>Bathynectes longipes</i> , <i>Bathynectes maravigna</i> , <i>Brachynotus sexdentatus</i> , <i>Calappa granulata</i> , <i>Calappa pelii</i> , <i>Calappa tuerkayana</i> , <i>Calcinus tubularis</i> , <i>Dardanus arrosor</i> , <i>Dardanus calidus</i> , <i>Derilambrus angulifrons</i> , <i>Dicranodromia mahieuxii</i> , <i>Dorhynchus thomsoni</i> , <i>Dromia personata</i> , <i>Ebalia cranchii</i> , <i>Ebalia deshayesi</i> , <i>Ebalia granulosa</i> , <i>Ebalia nux</i> , <i>Ebalia tuberosa</i> , <i>Ergasticus clouei</i> , <i>Eriphia verrucosa</i> , <i>Erugosquilla massavensis</i> , <i>Ethusa mascarone</i> , <i>Eurynome aspera</i> , <i>Galathea dispersa</i> , <i>Galathea intermedia</i> , <i>Galathea nexa</i> , <i>Galathea strigosa</i> , <i>Geryon longipes</i> , <i>Geryon trispinosus</i> , <i>Goneplax rhomboides</i> , <i>Homola barbata</i> , <i>Ilia nucleus</i> , <i>Inachus aguiarii</i> , <i>Inachus communissimus</i> , <i>Inachus dorsettensis</i> , <i>Inachus leptochirus</i> , <i>Inachus parvirostris</i> , <i>Inachus phalangium</i> , <i>Inachus thoracicus</i> , <i>Jaxea nocturna</i> , <i>Latreillia elegans</i> , <i>Liocarcinus corrugatus</i> , <i>Liocarcinus depurator</i> , <i>Liocarcinus maculatus</i> , <i>Liocarcinus navigator</i> , <i>Liocarcinus vernalis</i> , <i>Lissa chiragra</i> , <i>Macropipus tuberculatus</i> , <i>Macropodia linareti</i> , <i>Macropodia longipes</i> , <i>Macropodia longirostris</i> , <i>Macropodia rostrata</i> , <i>Macropodia tenuirostris</i> , <i>Maja brachydactyla</i> , <i>Maja crispata</i> , <i>Maja goletziana</i> , <i>Medorippe lanata</i> , <i>Monodaeus couchii</i> , <i>Munida curvimana</i> , <i>Munida intermedia</i> , <i>Munida rugosa</i> , <i>Munida rutilanti</i> , <i>Munida tenuimana</i> , <i>Nepinnotheres pinnotheres</i> , <i>Paguristes eremita</i> , <i>Pagurus alatus</i> , <i>Pagurus anachoretus</i> , <i>Pagurus cuanensis</i> , <i>Pagurus excavates</i> , <i>Pagurus forbesii</i> , <i>Pagurus prideaux</i> , <i>Palicus caronii</i> , <i>Paractaea monodi</i> , <i>Parasquilla ferussaci</i> , <i>Paromola cuvieri</i> , <i>Parthenope expansa</i> , <i>Parthenopidae</i> , <i>Parthenopoides Massena</i> , <i>Pilumnus hirtellus</i> , <i>Pilumnus spinifer</i> , <i>Pilumnus villosissimus</i> , <i>Pinnotheres pisum</i> , <i>Pisa armata</i> , <i>Pisa nodipes</i> , <i>Pisidia longicornis</i> , <i>Polybius henslowii</i> , <i>Polychelus typhlops</i> , <i>Portumnus latipes</i> , <i>Portunus hastatus</i> , <i>Pseudosquillaopsis cerisii</i> , <i>Rissoides desmaresti</i> , <i>Rissoides pallidus</i> , <i>Rochinia carpenter</i> , <i>Scyllarides latus</i> , <i>Scyllarus arctus</i> , <i>Scyllarus pygmaeus</i> , <i>Sphaerozium nitidus</i> , <i>Spinolambrus macrochelous</i> , <i>Thalamita poissonii</i> , <i>Thia scutellata</i> , <i>Xantho pilipes</i>		
B		International Bottom Trawl Survey in the Mediterranean (Medits)
P/B	$\log(P/B) = 10.154 - 0.271 * \log(M) - 2824.247 * (1/(T+273)) - 0.063 * \log(D+1) + 0.13 * (Me) + 0.076 * (Car) - 0.311 * (Mol) - 0.154 * (Crus) - 0.266 * (Pol) - 0.472 * (Echi)$	603,604,647
Q/B		583
D		630,632,648-652
C		GFCM-FAO
Di		306,312,313,315,323-327,329,339,377,378,389,390,401,496,534
61. Mobile benthos: <i>Amphipholis squamata</i> , <i>Amphipoda</i> , <i>Amphiura chiajei</i> , <i>Amphiura</i> spp., <i>Annelida</i> , <i>Anseropoda placenta</i> , <i>Antalis entalis</i> , <i>Antedon mediterranea</i> , <i>Antedon</i> spp., <i>Aphrodita aculeata</i> , <i>Aphrodita</i> spp., <i>Aspidosiphon muelleri</i> muelleri, <i>Asteroidea</i> , <i>Astropecten aranciatus</i> , <i>Astropecten bispinosus</i> , <i>Astropecten irregularis pentacanthus</i> , <i>Astropecten jonstoni</i> , <i>Astropecten spinulosus</i> , <i>Astropecten</i> spp., <i>Astrosparitus mediterraneus</i> , <i>Bonelliidae</i> , <i>Brissopsis atlantica</i> , <i>Brissopsis atlantica mediterranea</i> , <i>Brissopsis lyrifera</i> , <i>Brissus unicolor</i> , <i>Centrostephanus longispinus</i> , <i>Ceramaster grenadensis</i> , <i>Chaetaster longipes</i> , <i>Chloeia venusta</i> , <i>Cidaris cidaris</i> , <i>Coscinasterias tenuispina</i> , <i>Dentaliidae</i> , <i>Diopatra neapolitana</i> , <i>Echinaster sepositus</i> , <i>Echinidae</i> , <i>Echinocardium cordatum</i> , <i>Echinocardium mediterraneum</i> , <i>Echinus melo</i> , <i>Euphrosine foliosa</i> , <i>Filograna implexa</i> , <i>Genocidaris maculata</i> , <i>Golfingia vulgaris vulgaris</i> , <i>Gracilechinus acutus</i> , <i>Hacelia attenuata</i> , <i>Havelockia inermis</i> , <i>Hediste diversicolor</i> , <i>Hirudinea</i> , <i>Holothuria forskali</i> , <i>Holothuria helleri</i> , <i>Holothuria mammata</i> , <i>Holothuria poli</i> , <i>Holothuria</i> spp., <i>Holothuria tubulosa</i> , <i>Hyalinoecia tubicola</i> , <i>Hymenodiscus coronata</i> , <i>Hyperiididae</i> , <i>Idotea</i> spp., <i>Isopoda</i> , <i>Labidoplax digitata</i> , <i>Laetmonice hystrix</i> , <i>Laetmonice</i> spp., <i>Leptometra phalangium</i> , <i>Leptometra</i> spp., <i>Leptopentacta elongate</i> , <i>Leptopentacta</i> spp., <i>Leptopentacta tergestina</i> , <i>Luidia ciliaris</i> , <i>Luidia sarsii</i> , <i>Luidia</i> spp., <i>Lumbrineris</i> spp., <i>Marthasterias glacialis</i> , <i>Mesothuria intestinalis</i> , <i>Molpadia musculus</i> , <i>Molpadia</i> spp., <i>Natatolana borealis</i> , <i>Ocnus planci</i> , <i>Ocnus syracusanus</i> , <i>Odontaster mediterraneus</i> , <i>Ophiacantha setosa</i> , <i>Ophidiaster ophidianus</i> , <i>Ophiocomina nigra</i> , <i>Ophioderma longicauda</i> , <i>Ophiomyxa pentagona</i> , <i>Ophiopsila aranea</i> , <i>Ophiorthrix fragilis</i> , <i>Ophiorthrix quinqueaculata</i> , <i>Ophiorthrix</i> spp., <i>Ophiura albida</i> , <i>Ophiura ophiura</i> , <i>Ophiura</i> spp., <i>Ophiuridae</i> , <i>Ophiuroidea</i> , <i>Ova canaliferus</i> , <i>Oweniidae</i> , <i>Paracentrotus lividus</i> , <i>Parastichopus regalis</i> , <i>Peltaster placenta</i> , <i>Phronima sedentaria</i> , <i>Phronima</i> spp., <i>Phrosina semilunata</i> , <i>Phyllophorus urna</i> , <i>Plutonaster bifrons</i> , <i>Polychaeta</i> , <i>Pontobdella muricata</i> , <i>Priapulidae</i> , <i>Protula intestinum</i> , <i>Psammecinus microtuberculatus</i> , <i>Sipunculidae</i> , <i>Sipunculus nudus</i> , <i>Sipunculus</i> spp., <i>Spatangus purpureus</i> , <i>Spatangus subinermis</i> , <i>Sphaerechinus granularis</i> , <i>Sternaspis scutata</i> , <i>Stylocidaris affinis</i> , <i>Synaptidae</i> , <i>Tubulanus</i> spp.		
B		International Bottom Trawl Survey in the Mediterranean (Medits)

P/B	$\log(P/B) = 10.154 - 0.271 * \log(M) - 2824.247 * (1/(T+273)) - 0.063 * \log(D+1) + 0.13 * (Me) + 0.076 * (Car) - 0.311 * (Mol) - 0.154 * (Crus) - 0.266 * (Pol) - 0.472 * (Echi)$	603,604
Q/B		583
D		653-657
Di		247,312,313,324,329,339,534
62. Sessile benthos: <i>Actinauge richardi</i> , <i>Actinauge</i> spp., <i>Actinia</i> spp., <i>Actiniidae</i> , <i>Adamsia palliata</i> , <i>Adamsia</i> spp., <i>Agelas oroides</i> , <i>Amathia semiconvoluta</i> , <i>Andresia parthenopea</i> , <i>Aplidium conicum</i> , <i>Aplidium elegans</i> , <i>Aplidium haouarianum</i> , <i>Aplidium nordmanni</i> , <i>Aplidium pallidum</i> , <i>Aplidium proliferum</i> , <i>Aplidium</i> spp., <i>Ascidia involuta</i> , <i>Ascidia mentula</i> , <i>Ascidia</i> spp., <i>Ascidia virginea</i> , <i>Asciidiella aspersa</i> , <i>Asciidiella scabra</i> , <i>Asciidiella</i> spp., <i>Asciidiidae</i> , <i>Aulactinia verrucosa</i> , <i>Axinella cannabina</i> , <i>Axinella damicornis</i> , <i>Axinella polypoides</i> , <i>Axinella</i> spp., <i>Axinella verrucosa</i> , <i>Beania cylindrica</i> , <i>Botryllus schlosseri</i> , <i>Botryllus</i> spp., <i>Bryozoa</i> , <i>Bugula neritina</i> , <i>Bugula</i> spp., <i>Calliactis parasitica</i> , <i>Calpensia nobilis</i> , <i>Caryophyllia smithii</i> , <i>Caryophyllia</i> spp., <i>Cellaria salicornioides</i> , <i>Celleporina caliciformis</i> , <i>Cerianthus membranaceus</i> , <i>Cerianthus</i> spp., <i>Chondrosia reniformis</i> , <i>Ciona edwardsi</i> , <i>Ciona intestinalis</i> , <i>Ciona</i> spp., <i>Clathria coralloides</i> , <i>Cystodytes dellechiaiei</i> , <i>Diazona violacea</i> , <i>Didemnidae</i> , <i>Didemnum maculosum</i> , <i>Didemnum</i> spp., <i>Diplosoma listerianum</i> , <i>Distaplia magnilarva</i> , <i>Distomus variolosus</i> , <i>Epizoanthus arenaceus</i> , <i>Epizoanthus paguricola</i> , <i>Epizoanthus</i> spp., <i>Eudistoma mucosum</i> , <i>Eudistoma</i> spp., <i>Filicrisia</i> spp., <i>Fron dipora verrucosa</i> , <i>Funiculina quadrangularis</i> , <i>Geodia cydonium</i> , <i>Geodia</i> spp., <i>Halecium halecinum</i> , <i>Haliclona simulans</i> , <i>Halocyntia papillosa</i> , <i>Hippospongia communis</i> , <i>Hormathia alba</i> , <i>Hydractinia echinata</i> , <i>Hydrozoa</i> , <i>Ircinia</i> spp., <i>Lytocarpia myriophyllum</i> , <i>Metridium senile</i> , <i>Microcosmus claudicans</i> , <i>Microcosmus polymorphus</i> , <i>Microcosmus sabatieri</i> , <i>Microcosmus</i> spp., <i>Microcosmus squamiger</i> , <i>Microcosmus vulgaris</i> , <i>Molgula appendiculata</i> , <i>Molgula occulta</i> , <i>Molgula socialis</i> , <i>Molgula</i> spp., <i>Myriapora truncata</i> , <i>Nemertesia antennina</i> , <i>Nemertesia ramosa</i> , <i>Nemertesia</i> spp., <i>Parazoanthus axinellae</i> , <i>Pennatula aculeata</i> , <i>Pennatula phosphorea</i> , <i>Pennatula rubra</i> , <i>Pennatula</i> spp., <i>Pennatulidae</i> , <i>Pentapora fascialis fascialis</i> , <i>Pentapora foliacea</i> , <i>Petrosia ficiformis</i> , <i>Phallusia mammillata</i> , <i>Poecillastra compressa</i> , <i>Polycarpa mamillaris</i> , <i>Polycarpa pomaria</i> , <i>Polycarpa</i> spp., <i>Polycitor adriaticus</i> , <i>Polycitor crystallinus</i> , <i>Polycitor</i> spp., <i>Polyclinella azemai</i> , <i>Polyclinidae</i> , <i>Porifera</i> , <i>Protula tubularia</i> , <i>Pseudodistoma cyrnusense</i> , <i>Pteroeides griseum</i> , <i>Pteroeides spinosum</i> , <i>Pteroeides</i> spp., <i>Pyura dura</i> , <i>Pyura microcosmus</i> , <i>Pyura</i> spp., <i>Pyura tessellata</i> , <i>Reteporella beaniana</i> , <i>Reteporella grimaldii</i> , <i>Reteporella</i> spp., <i>Rhizaxinella pyrifer</i> , <i>Sabella spallanzanii</i> , <i>Sabella</i> spp., <i>Sabellidae</i> , <i>Sagartia elegans</i> , <i>Sarcotragus foetidus</i> , <i>Scalarispongia scalaris</i> , <i>Scalpellum scalpellum</i> , <i>Schizobrachiella sanguinea</i> , <i>Scrupocellaria scrupea</i> , <i>Serpula vermicularis</i> , <i>Serpulidae</i> , <i>Sertularella crassicaulis</i> , <i>Sertularella</i> spp., <i>Spongia officinalis</i> , <i>Spongiidae</i> , <i>Styela canopus</i> , <i>Styela</i> spp., <i>Suberites carnosus</i> , <i>Suberites domuncula</i> , <i>Suberites ficus</i> , <i>Suberites</i> spp., <i>Synoicum blochmanni</i> , <i>Telmatactis forskali</i> , <i>Tethya aurantium</i> , <i>Tethya citrina</i> , <i>Tethyaster subinermis</i> , <i>Thenea muricata</i> , <i>Veretillidae</i> , <i>Veretillum cynomorium</i> , <i>Veretillum</i> spp., <i>Virgularia mirabilis</i>		
B		International Bottom Trawl Survey in the Mediterranean (MeditS)
P/B	$\log(P/B) = 10.154 - 0.271 * \log(M) - 2824.247 * (1/(T+273)) - 0.063 * \log(D+1) + 0.13 * (Me) + 0.076 * (Car) - 0.311 * (Mol) - 0.154 * (Crus) - 0.266 * (Pol) - 0.472 * (Echi)$	603,604
Q/B		583
D		658-660
Di		312,324,329,339,534
63. Jellyfish&Salps: <i>Aequorea forskalea</i> , <i>Aurelia aurita</i> , <i>Chrysaora hysoscella</i> , <i>Cnidaria</i> , <i>Pelagia noctiluca</i> , <i>Pleurobrachia pileus</i> , <i>Pyrosoma</i> , <i>Pyrosoma atlanticum</i> , <i>Rhizostoma pulmo</i> , <i>Thalia democratica</i> , <i>Salpa maxima</i> , <i>Salpa</i> spp.		
B		661
P/B	M/K	662
Q/B	(P/B)/(P/Q)	
D		663-669
Di		313,324,339,389,670,671
64. Corals&gorgonians: <i>Acanthogorgia hirsuta</i> , <i>Alcyonidium</i> spp., <i>Alcyonium acaule</i> , <i>Alcyonium palmatum</i> , <i>Alcyonium</i> spp., <i>Antipathella subpinnata</i> , <i>Antipathes dichotoma</i> , <i>Bebyrce mollis</i> , <i>Callogorgia verticillata</i> , <i>Corallium rubrum</i> , <i>Desmophyllum dianthus</i> , <i>Eunicella cavolini</i> , <i>Eunicella filiformis</i> , <i>Eunicella singularis</i> , <i>Eunicella</i> spp., <i>Eunicella verrucosa</i> , <i>Funiculina quadrangularis</i> , <i>Gorgoniidae</i> , <i>Isidella</i>		

<i>elongata, Kophobelemnon stelliiferum, Leptogorgia sarmentosa, Lophelia pertusa, Madrepora oculata, Paramuricea macrospina, Paramuricea clavata, Parantipathes larix, Pennatula spp., Swiftia pallida, Villogorgia bebyricoides, Viminella flagellum</i>		
B		672-688,689; International Bottom Trawl Survey in the Mediterranean (Medit)
P/B		672,675,677,685,686,690,691
Q/B	(P/B)/(P/Q)	
D		692-697
C		GFCM-FAO
Di		324,698
65. Zooplankton		
B		699
P/B		699
Q/B		699
D		699
66. Seagrass: Cymodocea nodosa, Posidonia oceanica, Zoostera marina, Zoostera noltii		
B		700-741
P/B		704,708,718,742-746
Di		312,725
67. Seaweed: Cystoseira barbata, Cystoseira brachycarpa balearica, Cystoseira cf. Elegans, Cystoseira compressa, Cystoseira compressa pustulata, Cystoseira corniculata, Cystoseira foeniculacea latiramosa, Cystoseira foeniculacea tenuiramosa, Cystoseira spinosa, Sargassum vulgare, Cutleria multifida, Hildenbrandia crouaniorum, Lobophora variegata, Palmophyllum crassum, Peyssonnelia rosa-marina, Polystrata fosliei, Pseudolithoderma adriaticum, Corallinaceae encrusting, Lithophyllum sp., Lithophyllum stictaeforme, Mesophyllum alternans, Acrodiscus vidovichii, Acrosymphyton purpuriferum, Amphiroa cryptarthrodia, Amphiroa rigida, Bonnemaisonia asparagoides, Caulerpa prolifera, Codium bursa, Codium cf. Coralloides, Codium effusum, Codium vermilara, Colpomenia sinuosa, Cryptonemia lomation, Dictyopteris polypodioides, Dictyota dichotoma, Dictyota fasciola, Dictyota implexa, Digenea simplex, Dudresnaya verticillata, Flabellia petiolata, Halimeda tuna, Halopteris filicina, Halopteris scoparia, Hydroclathrus clathratus, Liagora viscida, Nereia filiformis, Padina pavonica, Peyssonnelia harveyana, Peyssonnelia squamaria, Rytiphlaea tinctoria, Scinaia furcellata, Scinaia sp., Sebdenia sp., Sphaerococcus coronopifolius, Stilophora tenella, Taonia atomaria, Tricleocarpa fragilis, Ulva rigida, Zanardinia typus, Acrothamnion preissii, Asparagopsis armata, Asparagopsis taxiformis, Caulerpa racemosa cylindracea, Caulerpa racemosa lamourouxii, Lophocladia lallemandii, Stypopodium schimperi, Womersleyella setacea, Acetabularia acetabulum, Acinetospora crinita, Acrosorium ciliolatum, Alsidium corallinum, Anadyomene stellate, Asperococcus bullosus, Botryocladia botryoides, Botryocladia chiajeana, Bryopsis sp., Chaetomorpha sp., Champia parvula, Chondracanthus acicularis, Chondrophycus sp., Cladophora pellucida, Cladophora prolifera, Cladophora sp., Cladostephus spongiosus, Corallina elongate, Corallinaceae articulated, Cyanobacteria, Dasycladus vermicularis, Elachista intermedia, Gelidium bipectinatum, Gigartinales unidentified, Halitilon virgatum, Irvinea boergesenii, Laurencia gr. Obtusa, Laurencia sp., Parvocaulis parvulus, Pseudochlorodesmis furcellata, Rhodymenia ardissonae, Sphacelaria cirrosa, Valonia utricularis, Wrangelia penicillata		
B		747-757
P/B		753
68. Small phytoplankton: Dinoflagellates		
B		699
P/B		699

B		699
P/B		699
B		699
Fisheries: Beach Seine, Dredge, Driftnet, Gillnet, Longline, Midwater Trawl, Purse Seine, Trammel net, Trap&Pot, Trawler		
Catch/Di		GFCM-FAO; DCF (EC)
Effort		GFCM-FAO; DCF (EC)

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Modelling the Mediterranean Sea ecosystem at high spatial resolution to inform the ecosystem-based management in the region

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S4.

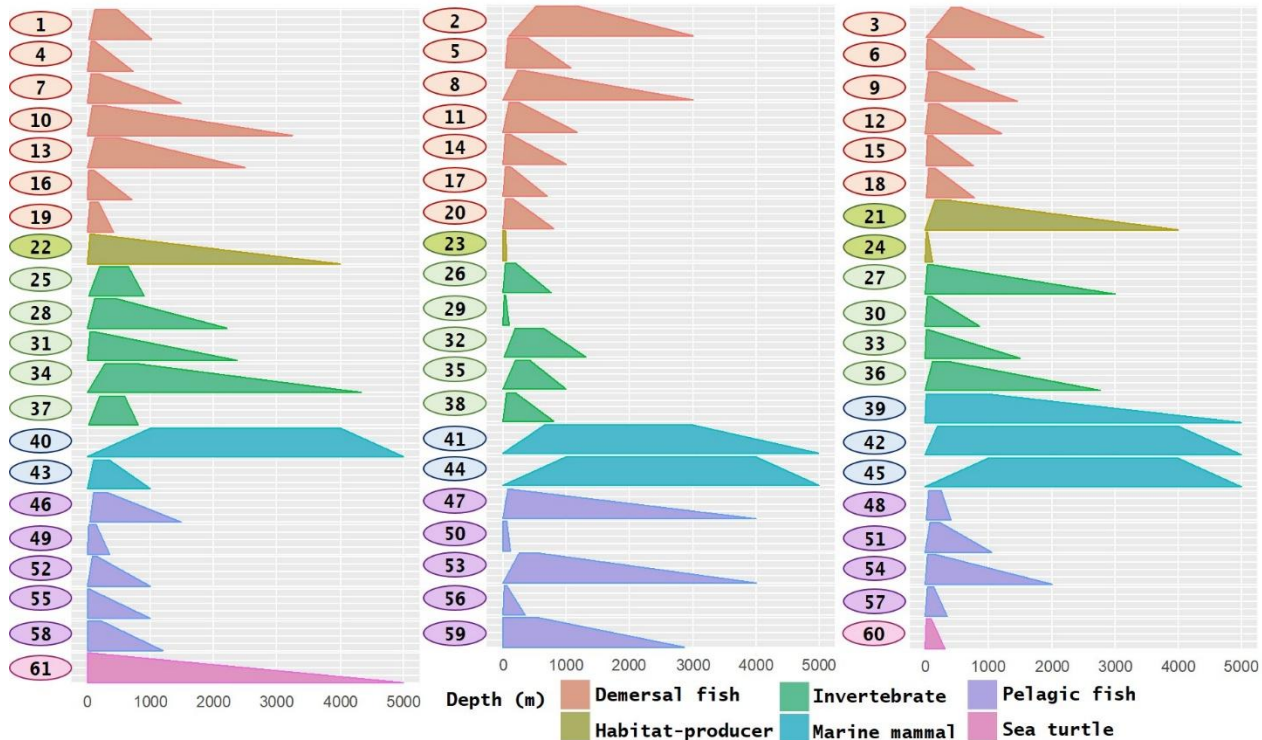


Fig S2. Depth response curves used for selected species/functional groups of the Mediterranean Sea ecosystem. **Demersal fish:** 1: Anglerfish; 2: Bathydemersal fish; 3: Blackmouth catshark; 4: Other commercial small demersal fish; 5: European hake (Adult); 6: European hake (Recruit); 7: Flatfish; 8: Gadidae; 9: Non-commercial medium demersal fish; 10: Non-commercial small demersal fish; 11: Other commercial large demersal fish; 12: Other commercial medium demersal fish; 13: Other small demersal shark; 14: Ray&Skate; 15: Red mullet (Adult); 16: Red mullet (Recruit); 17: Sparidae; 18: Small-spotted catshark; 19: Surmullet; 20: Torpedo; **Habitat-producer:** 21: Coral&Gorgonian; 22: Sessile benthos; 23: Seagrass; 24: Seaweed; **Invertebrates:** 25: Blue and red shrimp; 26: Benthic cephalopod; 27: Bivalve; 28: Benthopelagic cephalopod; 29: Commercial prawn; ; 30: Deep-water rose shrimp; 31: Gastropod; 32: Giant red shrimp; 33: Mobile benthos; 34: Mesopelagic cephalopod; 35: Non-commercial shrimp/prawn; 36: Non-commercial decapod; 37: Norway lobster; 38: Other commercial decapod; **Marine mammal:** 39: Bottlenose dolphin; 40: Common dolphin; 41: Deep Sea cetacean; 42: Fin whale; 43: Monk seal; 44: Striped dolphin; 45: Sperm whale; **Pelagic fish:** 46: Benthopelagic fish; 47: Bluefin tuna; 48: European anchovy (Adult); 49: European anchovy (Recruit); 50: European pilchard (Adult) and European pilchard (Recruit); 51: Horse mackerel; 52: Mackerel; 53: Mesopelagic fish; 54: Non-commercial large pelagic fish; 55: Other commercial large pelagic fish; 56: Other medium pelagic fish; 57: Other small pelagic fish; 58: Pelagic shark; 59: Swordfish; **Sea turtle:** 60: Green turtle; 61: Loggerhead turtle.

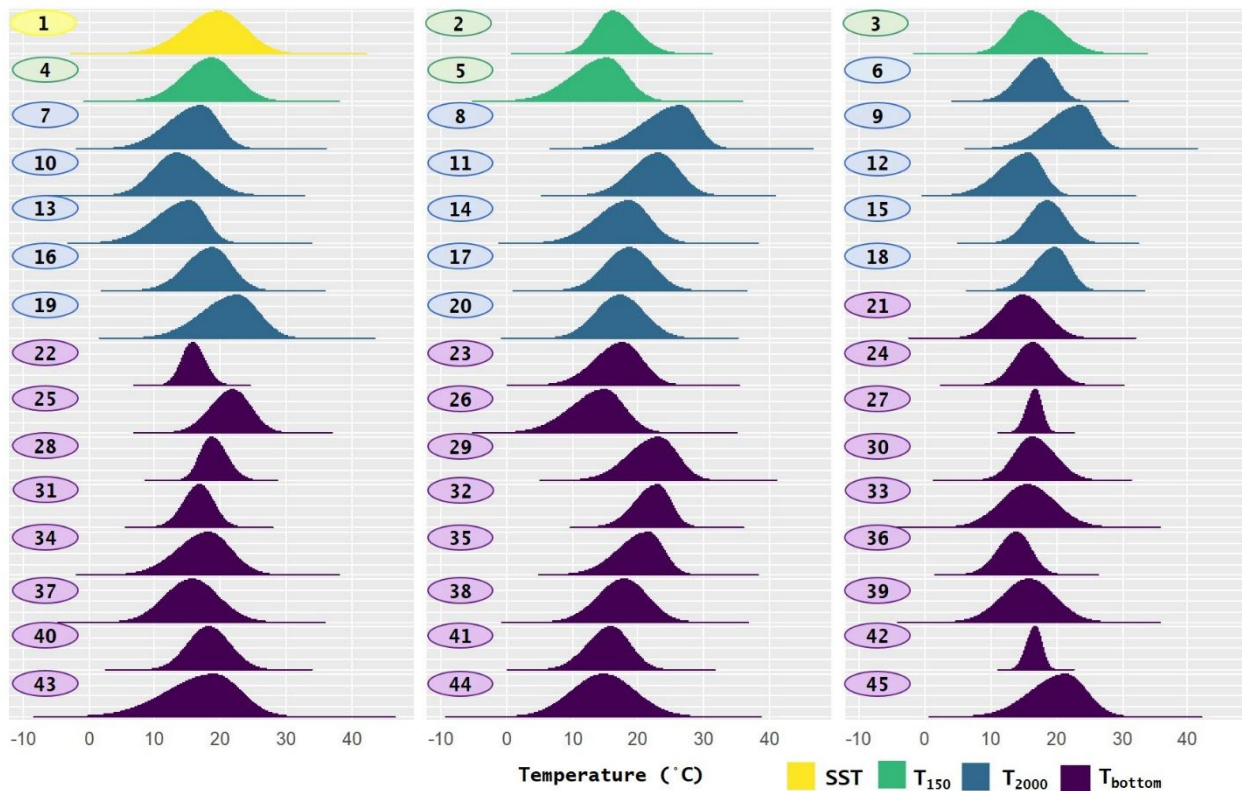


Fig S3. Temperature response curves used for selected species/functional groups of the Mediterranean Sea ecosystem. **SST:** 1: European anchovy (Adult) and European anchovy (Recruit); **T_{150} :** 2: Horse mackerel; 3: Mackerel; 4: Other medium pelagic fish; 5: European pilchard (Adult) and European pilchard (Recruit); **T_{2000} :** 6: Benthopelagic fish; 7: Benthopelagic cephalopod; 8: Loggerhead turtle; 9: Other commercial large pelagic fish; 10: Gadidae; 11: Green turtle; 12: European hake (Adult); 13: Mesopelagic fish; 14: Mesopelagic cephalopod; 15: Non-commercial medium demersal fish; 16: Other commercial medium demersal fish; 17: Pelagic shark; 18: Sparidae; 19: Swordfish; 20: Bluefin tuna; **T_{bottom} :** 21: Anglerfish; 22: Bathydemersal fish; 23: Benthic cephalopod; 24: Bivalve; 25: Blue and red shrimp; 26: Blackmouth catshark; 27: Coral&Gorgonian; 28: Other commercial small demersal fish; 29: Deep-water rose shrimp; 30: Flatfish; 31: Gastropod; 32: Giant red shrimp; 33: Non-commercial decapod; 34: Non-commercial small demersal fish; 35: Non-commercial shrimp/prawn; 36: Norway lobster; 37: Other commercial decapod; 38: Other commercial large demersal fish; 39: Other small demersal shark; 40: Ray&Skate; 41: Red mullet (Adult) and Red mullet (Recruit); 42: Sessile benthos; 43: Small-spotted catshark; 44: Surmullet; 45: Torpedo.

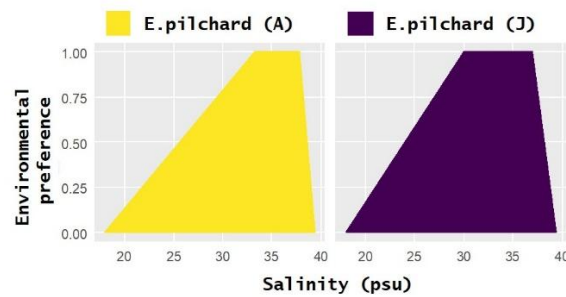


Figure S4: Salinity response curves used for European pilchard (Adult) and European pilchard (Recruit).

Other Ecospace input parameters

In Ecospace, the parameters that regulate spatial distribution of functional groups are: Base dispersal rate, Relative dispersal in bad habitat, and Relative vulnerability to predation in bad habitat (Table S4). Base Dispersal rate corresponds to swimming speed (km/year) and can be set between 0 and infinite; the default value is 300 km/year. The Relative dispersal in bad habitat increases dispersal rate in non-favourable habitats to simulate greater active attempt to move elsewhere with better conditions. It can be set from 1 (which inactivates the mechanism) to 10; the default is set at 5. Relative vulnerability to predation in bad habitat is a weight factor for the vulnerability parameter in Ecosim. It regulates the increased vulnerability to predation (or decreased sheltering capacity) in a less-than-optimal habitat. Its default value is 2 (twice more vulnerable in bad habitat), and it can be set between 1 and 100.

Table S4

#	Group name	Base dispersal rate	Relative dispersal in bad habitat (proportion)	Rel. vul to pred. in bad hab.
1	Bottlenose dolphin	300	5	2
2	Striped dolphin	300	5	2
3	Common dolphin	300	5	2
4	Fin whale	300	5	2
5	Sperm whale	300	5	2
6	Deep Sea cetaceans	300	5	2
7	Monk seal	300	5	2
8	Pelagic Seabird	300	5	2

9	Gull&Cormorant	300	5	2
10	Tern	200	5	2
11	Loggerhead turtle	300	5	2
12	Green turtle	300	5	2
13	Pelagic shark	300	5	2
14	NC large pelagic	300	5	2
15	Bluefin tuna	1000	5	2
16	Swordfish	300	5	2
17	Other large pelagic	300	5	2
18	Mackerel	300	5	2
19	Horse mackerel	200	5	10
20	Other medium pelagic	30	5	2
21	E. sardine Adult	30	5	2
22	E. sardine Recruit	30	5	2
23	E. anchovy Adult	30	5	2
24	E. Anchovy Recruit	30	5	2
25	Other small pelagic	30	5	2
26	Benthopelagic fish	30	5	2
27	Mesopelagic fish	30	5	2
28	Anglerfish	30	5	2
29	E. Hake Adult	30	5	2
30	E. Hake Recruit	30	5	2
31	OC large demersal	30	5	2
32	Gadidae	30	5	2
33	Sparidae	500	5	2
34	OC medium demersal f.	100	5	2
35	Flatfish	250	5	2
36	NC medium demersal f.	30	5	2
37	Red mullet Adult	80	5	2
38	Red mullet Recruit	80	5	2
39	Surmullet	80	5	2
40	OC small demersal f.	80	5	2

41	NC small demersal f.	60	5	10
42	Bathymodermis f.	30	5	2
43	Small-spotted catshark	30	5	2
44	Blackmouth catshark	300	5	10
45	Other small demersal shark	30	5	2
46	Ray&Skate	30	5	2
47	Torpedo	30	5	2
48	Benthic cephalopod	3	5	2
49	Benthopelagic cephalopod	300	5	2
50	Mesopelagic cephalopod	300	5	2
51	Bivalve	3	5	2
52	Gastropod	3	5	2
53	Deep-water rose shrimp	3	5	2
54	Blue and red shrimp	3	5	2
55	Giant red shrimp	3	5	2
56	Commercial prawn	3	5	2
57	NC shrimp/prawn	3	5	2
58	Norway lobster	3	5	2
59	OC decapod	3	5	2
60	NC decapod	3	5	2
61	Mobile benthos	3	5	2
62	Sessile benthos	1	5	2
63	Jellyfish&Salp	300	5	2
64	Coral&Gorgonian	1	5	2
65	Zooplankton	300	5	2
66	Seagrass	1	5	2
67	Seaweed	1	5	2
68	Small Phytoplankton	300	5	2
69	Large Phytoplankton	300	5	2
70	Discards	3	5	2
71	Detritus	1	5	2

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S5.

Table S5. Detailed description of modelled derived indicators with acronyms, definitions and references.

Ecological Indicator	Acronym	Definition and references
Total biomass	<i>TB</i>	Index calculated as the sum of the biomass of all the modelled functional groups (Fg; unit: t/km ²) ¹ .
Commercial biomass	<i>CB</i>	Index calculated as the sum of the biomass only for those Fg having a commercial value (unit: t/km ²) ² .
Invertebrates/Fish Biomass	<i>I/F</i>	Sum of the biomass of all invertebrate (commercial and non commercial) Fg / sum of the biomass of all (commercial and non commercial) fish Fg ^{3,4} . Explores processes benefiting the invertebrates or fish communities.
Demersal/Pelagic Biomass	<i>D/P</i>	Sum of the biomass of all demersal (commercial and non commercial) fish Fg/ sum of the biomass of all pelagic (commercial and non commercial) fish Fg ^{3,4} . Explores processes benefiting the demersal or pelagic compartments of the ecosystem.
Kempton species diversity index	<i>Qi</i>	The slope of the cumulative species abundance curve between the 10th and 90th percentiles. It expresses biomass species diversity and evenness by considering those Fg with trophic levels 3 or higher^{5,6}. The Kempton index is calculated as follow⁶: $Q_i = \frac{0.8Fg}{\log(\frac{R_2}{R_1})}$ where Fg is the total number of functional groups, R₁ and R₂ are the representative biomass values of the 10th and 90th percentiles in the cumulative abundance distribution.
Shannon entropy index	<i>H</i>	Shannon's entropy index⁷ is commonly used in ecological studies as a composite metric representing both species evenness and species richness. Applied to EwE however, where functional groups substitute for individual species and the number of functional groups is generally fixed, the index is useful for describing functional group evenness. Here, evenness represents the distribution of biomass across functional groups, where maximum evenness is achieved when all functional groups have equal biomass. The Shannon index is calculated as follow: $H = -\sum_{i=1}^n p_i \ln p_i$ where n represents the total number of functional groups and pi represents the proportion of total ecosystem biomass present in the ith functional group.
Mean trophic level of community	<i>TLco</i>	TL of the modelled community spans the whole ecosystem (living groups) ⁸ including all functional groups. TLs are calculated as follow ⁹ :

		$TL_j = 1 + \sum_{i=1}^n DC_{ji} \cdot TL_i$ where j is the predator of prey i, DC_{ji} is the fraction of prey i in the diet of each predator j, and TL_i is the TL of prey i. The mean trophic level of community is calculated as follow ¹⁰: $mTLco = \sum_{i=1}^n B_{Mi} \cdot TL_i / B_{MT}$ where B_{MT} is total biomass of the modeled ecosystem, B_{Mi} is the biomass of each species/Fg i in the model, and TL_i is the trophic level of species i as an output of the model (note: B_{Mi}, DC_{ji} and TL_i vary in time).
Mean trophic level of groups with TL >2	TL ₂	Calculated as the TLco, as the weighted mean (biomass) of the trophic level of species/Fg in the ecosystem with a trophic level ≥ 2 ¹⁰
Mean trophic level of groups with TL >3.25	TL _{3.25}	Calculated as the TLco, as the weighted mean (biomass) of the trophic level of species/Fg in the ecosystem with a trophic level ≥ 3.25 ¹⁰
Total Catch	TC	Sum of all catches (landings plus discards) of all Fg in the ecosystem (unit: t/km ² /year) ¹¹ .
Discards	Di	Sum of all catches of all Fg that are discarded (unit: t/km ² /year) ²
Trophic level of the catch	TLC	TL of the catch for all retained species/Fg. Retained species/Fg are caught in fishing operations, although not necessarily targeted by a fishery and which are retained because they are of commercial interest (i.e. not discarded) ⁸. Trophic level of the catch is calculated as follow ¹⁰: $TL_c = \sum_{i=1}^n Y_i \cdot TL_i / Y_L$ where Y_L is total landings, Y_i is the landing of species i and TL_i is the trophic level of species i (note: Y_L, Y_i and TL_i vary in time).

Table S6. List of species/functional groups defined included in the community indicators Demersal/Pelagic fish (D/P) and Invertebrates/Fish (I/F)

FG/Species	Organism	Ecology
13: Pelagic shark	Fish	Pelagic
14: Non commercial large pelagic	Fish	Pelagic
15: Bluefin tuna	Fish	Pelagic
16: Swordfish	Fish	Pelagic
17: Other large pelagic f.	Fish	Pelagic
18: Mackerel	Fish	Pelagic
19: Horse mackerel	Fish	Pelagic

20: Other medium pelagic f.	Fish	Pelagic
21 and 22: Sardine	Fish	Pelagic
23 and 24: Anchovy	Fish	Pelagic
25: Other small pelagic f.	Fish	Pelagic
26: Benthopelagic f.	Fish	Pelagic
27: Mesopelagic f.	Fish	Pelagic
28: Anglerfish	Fish	Demersal
29 and 30: Hake	Fish	Demersal
31: Other commercial large demersal f.	Fish	Demersal
32: Gadidae	Fish	Demersal
33: Sparidae	Fish	Demersal
34: Other commercial medium demersal f.	Fish	Demersal
35: Flatfish	Fish	Demersal
36: Non commercial medium demersal f.	Fish	Demersal
37 and 38: Mullet	Fish	Demersal
39: Surmullet	Fish	Demersal
40: Other commercial small demersal f.	Fish	Demersal
41: Non commercial small demersal f.	Fish	Demersal
42: Bathydemersal f.	Fish	Demersal
43: Small-spotted catshark	Fish	Demersal
44: Blackmouth catshark	Fish	Demersal
45: Other small demersal shark	Fish	Demersal
46: Ray&Skate	Fish	Demersal
47: Torpedo	Fish	Demersal
48: Benthic cephalopod	Invertebrates	Benthic
49: Benthopelagic cephalopod	Invertebrates	Benthopelagic
50: Mesopelagic cephalopod	Invertebrates	Mesopelagic
51: Bivalve	Invertebrates	Benthic
52: Gastropod	Invertebrates	Benthic
53: Deep-water rose shrimp	Invertebrates	Benthic
54: Blue and red shrimp	Invertebrates	Benthic

55: Giant red shrimp	Invertebrates	Benthic
56: Commercial prawn	Invertebrates	Benthic
57: Non commercial shrimp/prawn	Invertebrates	Benthic
58: Norway lobster	Invertebrates	Benthic
59: Other commercial decapod	Invertebrates	Benthic
60: Non commercial decapod	Invertebrates	Benthic
61: Mobile benthos	Invertebrates	Benthic
62: Sessile benthos	Invertebrates	Benthic
63: Jellyfish&Salp	Invertebrates	Pelagic
64: Coral&Gorgonian	Invertebrates	Benthic

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Modelling the Mediterranean Sea ecosystem at high spatial resolution to inform the ecosystem-based management in the region

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S6.

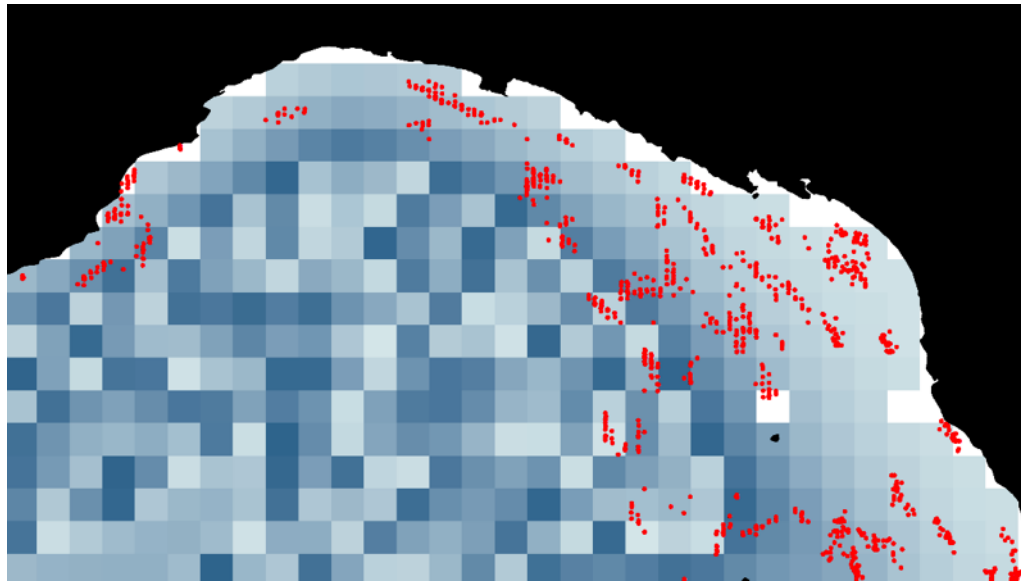


Fig S5. Example of data resolution of predicted model outputs (blue cells) vs Medits trawl survey (red dots).

Ecospace outputs:

Marine mammals, sea turtles and marine birds

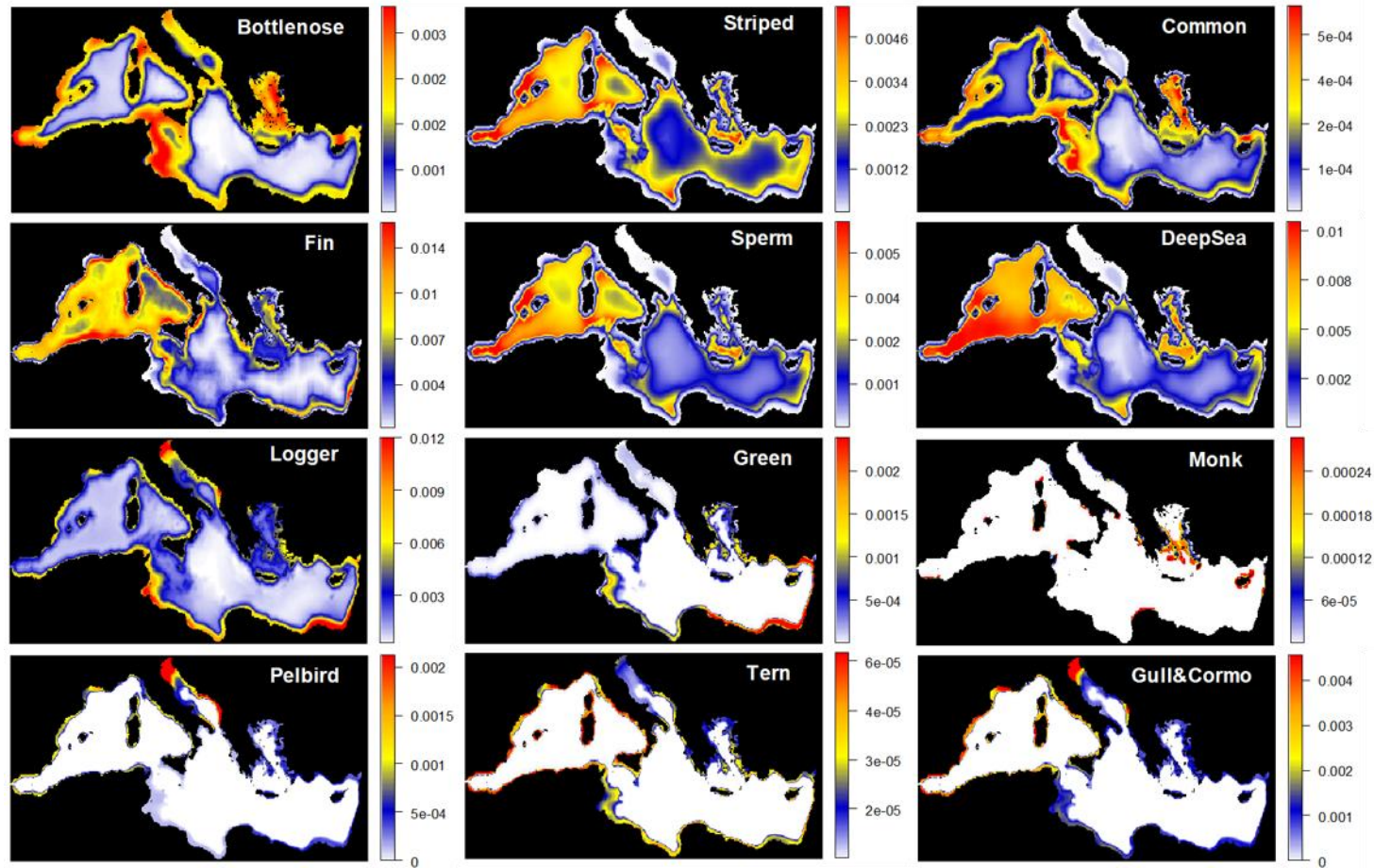


Fig S6. Mean (1995-2016) predicted biomass (t/km²) for marine mammals, sea turtles and marine birds. Bottlenose: Bottlenose dolphin; Striped: Striped dolphin; Common: Common dolphin; Fin: Fin whale; Sperm: Sperm whale; DeepSea: Deep sea cetaceans; Logger: Loggerhead turtle; Green: Green turtle; Monk: Monk seal; Pelbird: Pelagic bird; Gull&Cormo: Gull and Cormorant.

Pelagic fish

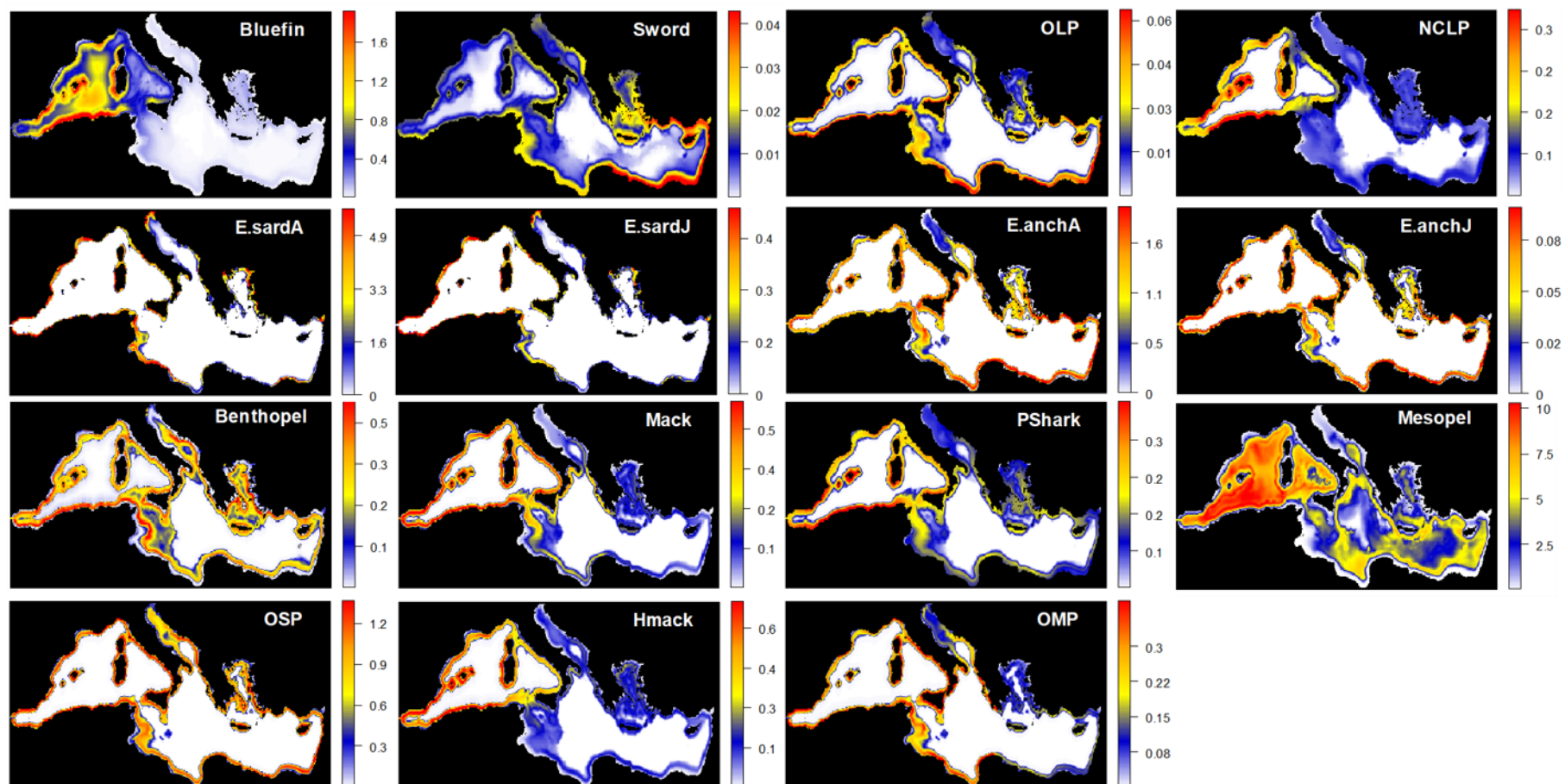


Fig S7. Mean (1995-2016) predicted biomass (t/km²) for pelagic fish groups. Bluefin: Bluefin tuna; Sword: Swordfish; OLP: Other large pelagic; NCLP: Non commercial large pelagic; E.sard: European pilchard; E.anch: European anchovy; Benthopel: Benthopelagic fish; Mack: Mackerel; PShark: Pelagic shark; Mesopel: Mesopelagic fish; OSP: Other small pelagic; Hmack: Horse mackerel; OMP: Other medium pelagic.

Demersal fish

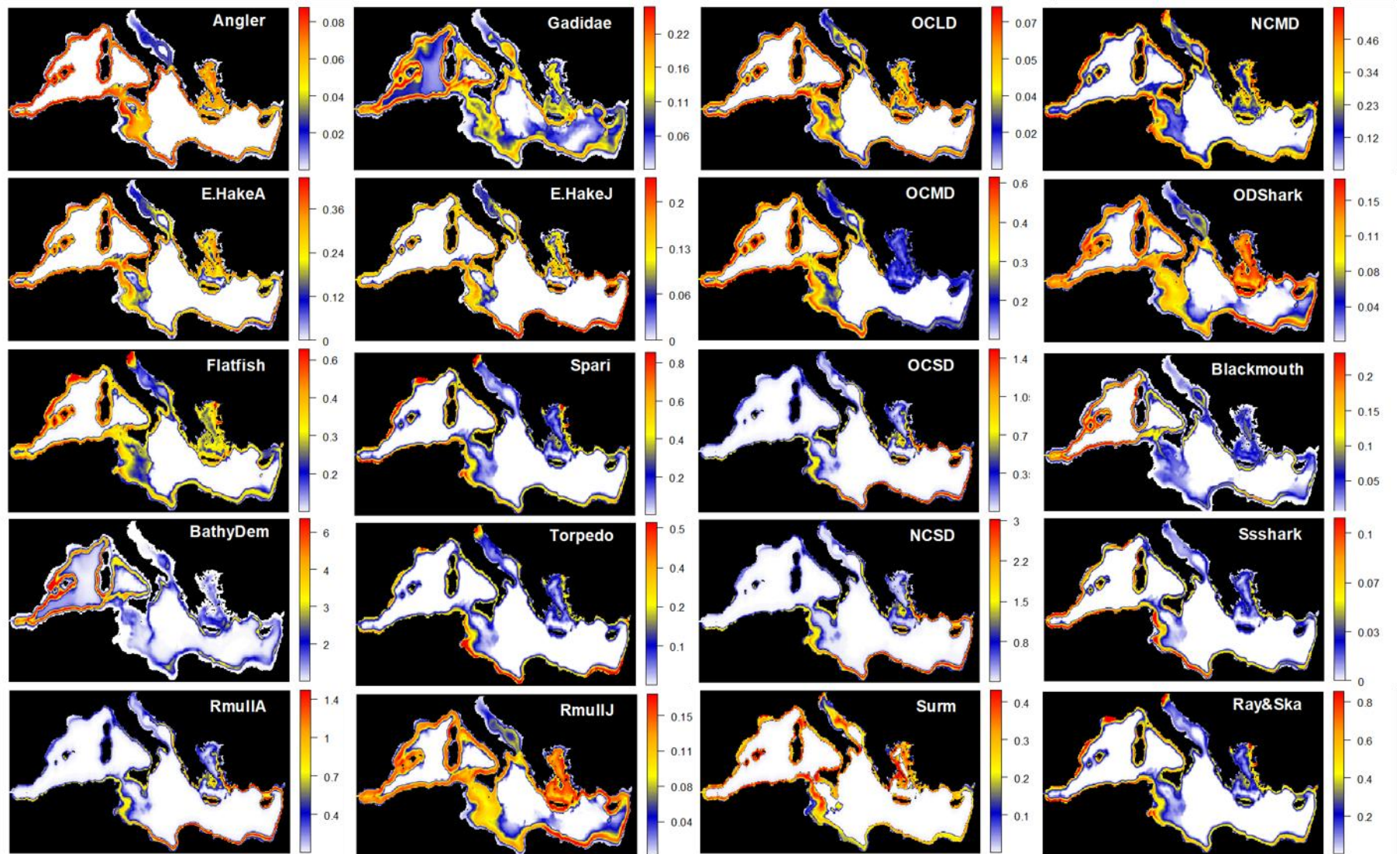


Fig S8. Mean (1995-2016) predicted biomass (t/km²) for demersal fish groups. OCLD: Other commercial large demersal; NCMD: Non-commercial medium demersal; OCMD: Other commercial medium demersal; ODshark: Other demersal sharks; Spari: Sparidae; OCSD: Other commercial small demersal; Blackmouth: Blackmouth catshark; BathyDem: Bathydemersal fish; NCSD: Non-commercial small demersal; Ssshark: Small-spotted catshark; Rmull: Red mullet; Surm: Surmullet; Ray&Ska: Ray&Skate.

Invertebrates

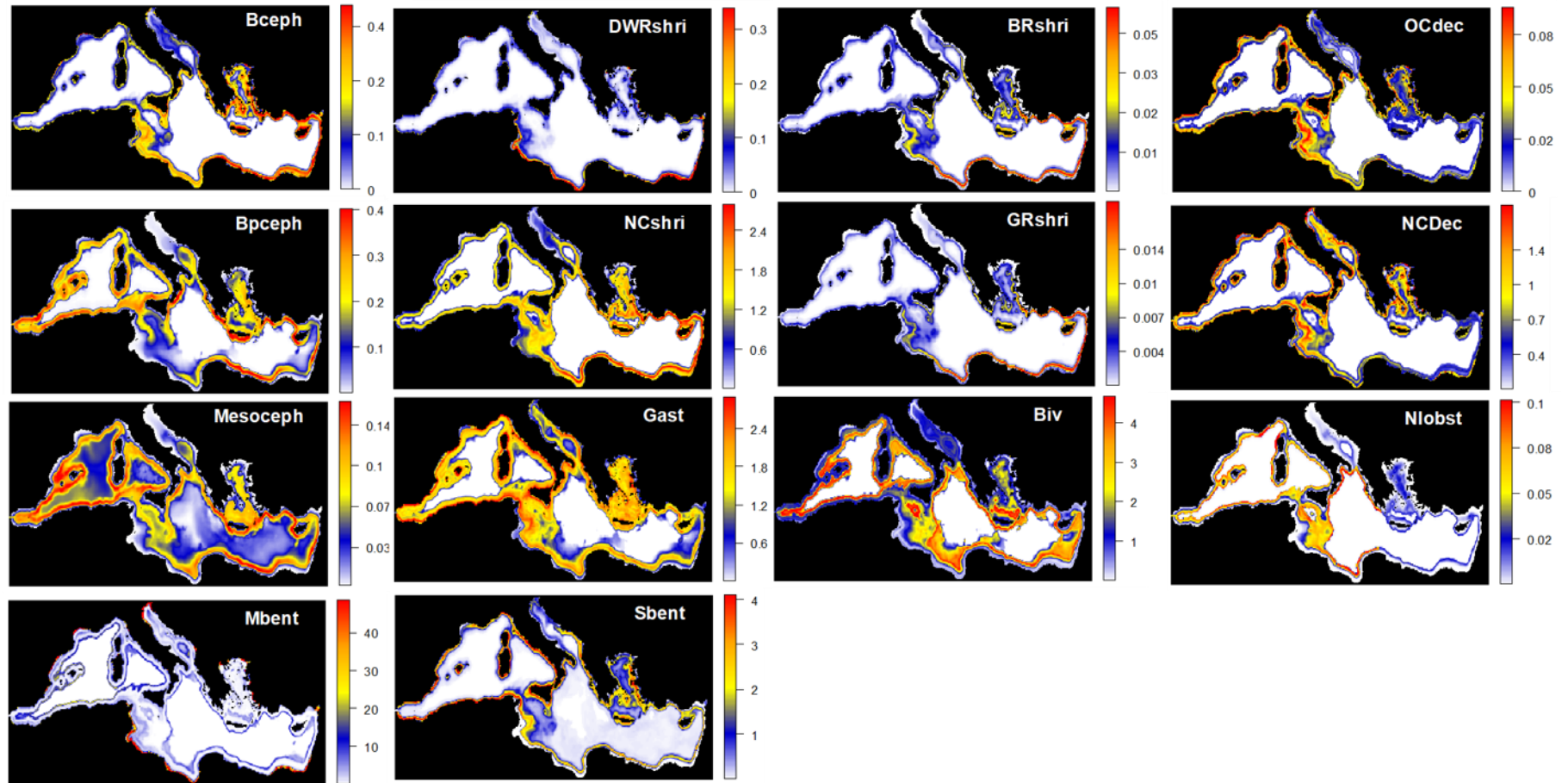


Fig S9. Mean (1995-2016) predicted biomass (t/km^2) for invertebrates groups. Bceph: Benthic cephalopod; DWRshri: Deep water rose shrimp; BRshri: Blue and red shrimp; OCdec: Other commercial decapod; Bpceph: Benthopelagic cephalopod; NCshri: Non commercial shrimp; GRshri: Giant red shrimp; NCDec: Non commercial decapod; Mesoceph: Mesopelagic cephalopod; Gast: Gastropod; Biv: Bivalve; Nlobst: Norway lobster; Mbent: Mobile benthos; Sbent: Sessile benthos.

Planktonic-Habitat

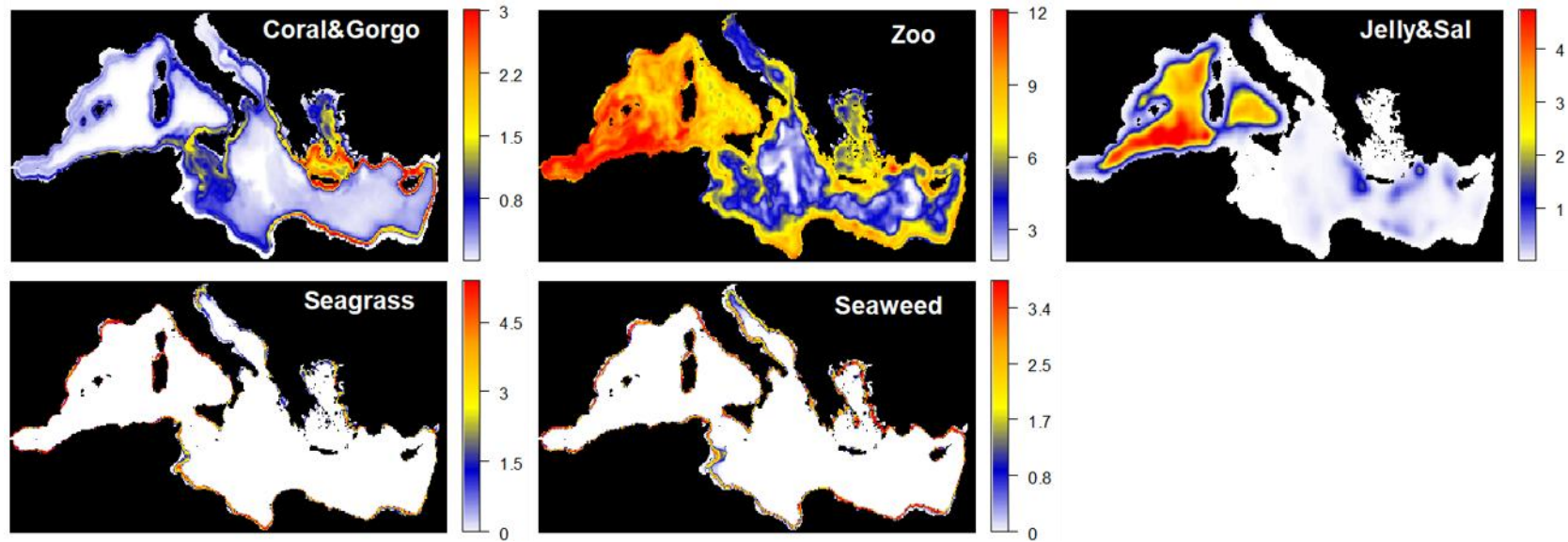


Fig S10. Mean (1995-2016) predicted biomass (t/km^2) for planktonic-habitat groups. Coral&Gorgo: Coral and gorgonian; Zoo: Zooplankton; Jelly&Sal: Jelly and Salp.

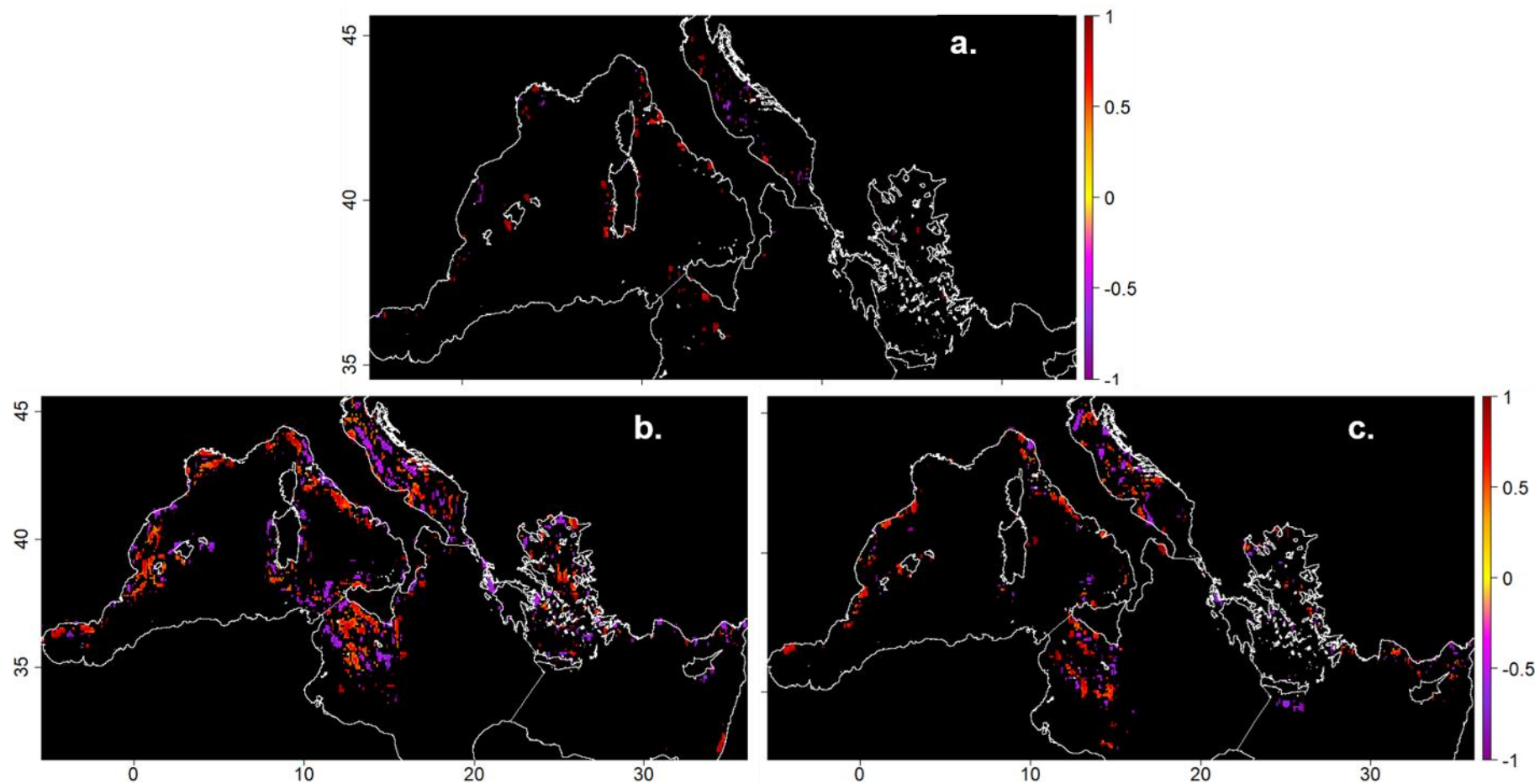


Fig S11. Mean (1995-2016) correlation between observed and predicted European hake (a.) and between AIS and predicted bottom trawlers (b) and purse seiners (c.) effort. Only correlations with p-value <0.05 are shown.

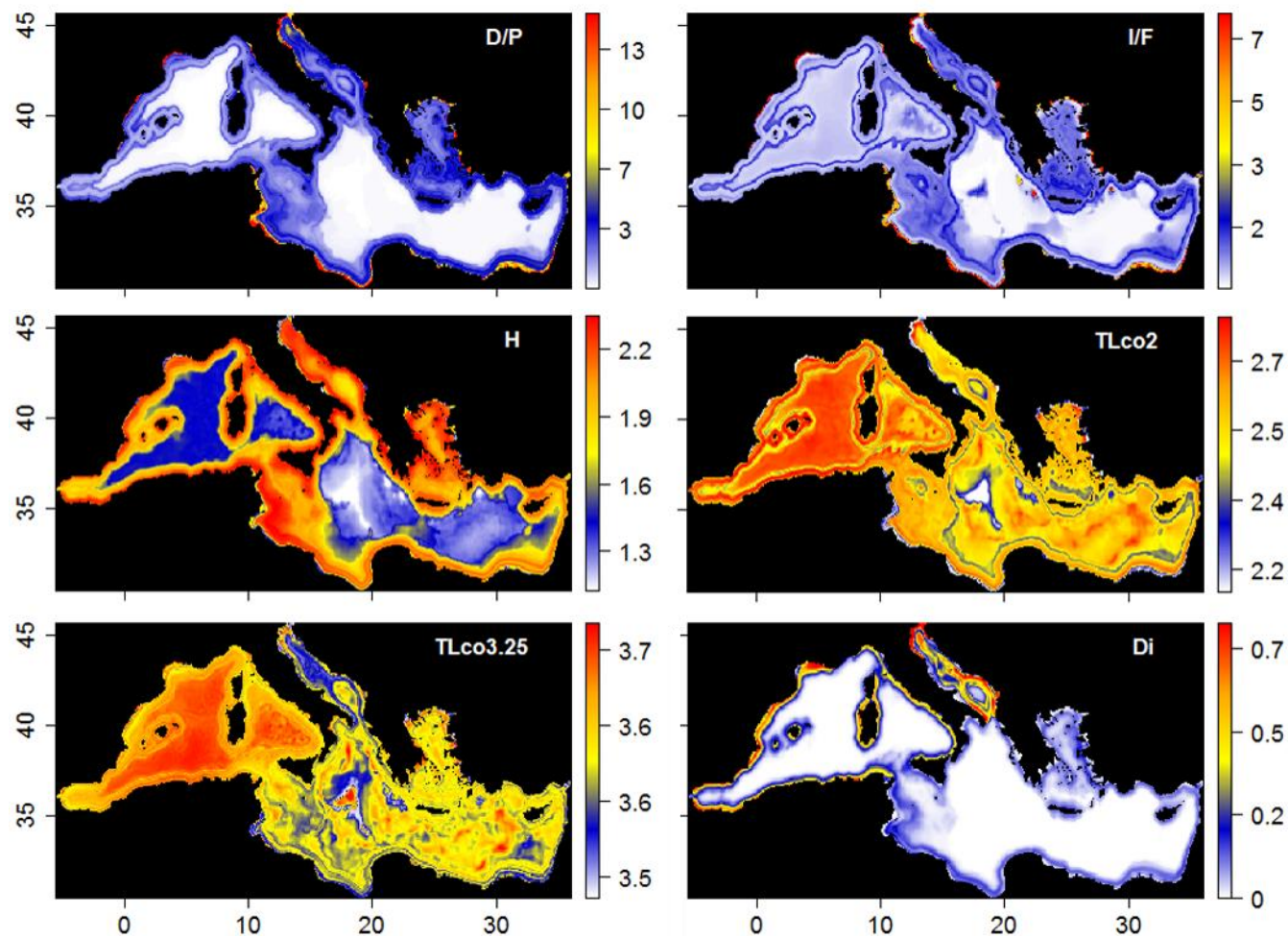


Fig S12. Mean (1995-2016) of selected modelled indicators. D/P: Demersal/Pelagic fish (t/km²); I/F: Invertebrates/Fish (t/km²); H: Shannon index; TLco2: Trophic level (TL) of species/Fg in the community with TL ≥ 2; Trophic level (TL) of species/Fg in the community with TL ≥ 3.25; Di: Discards (t/km²/year).

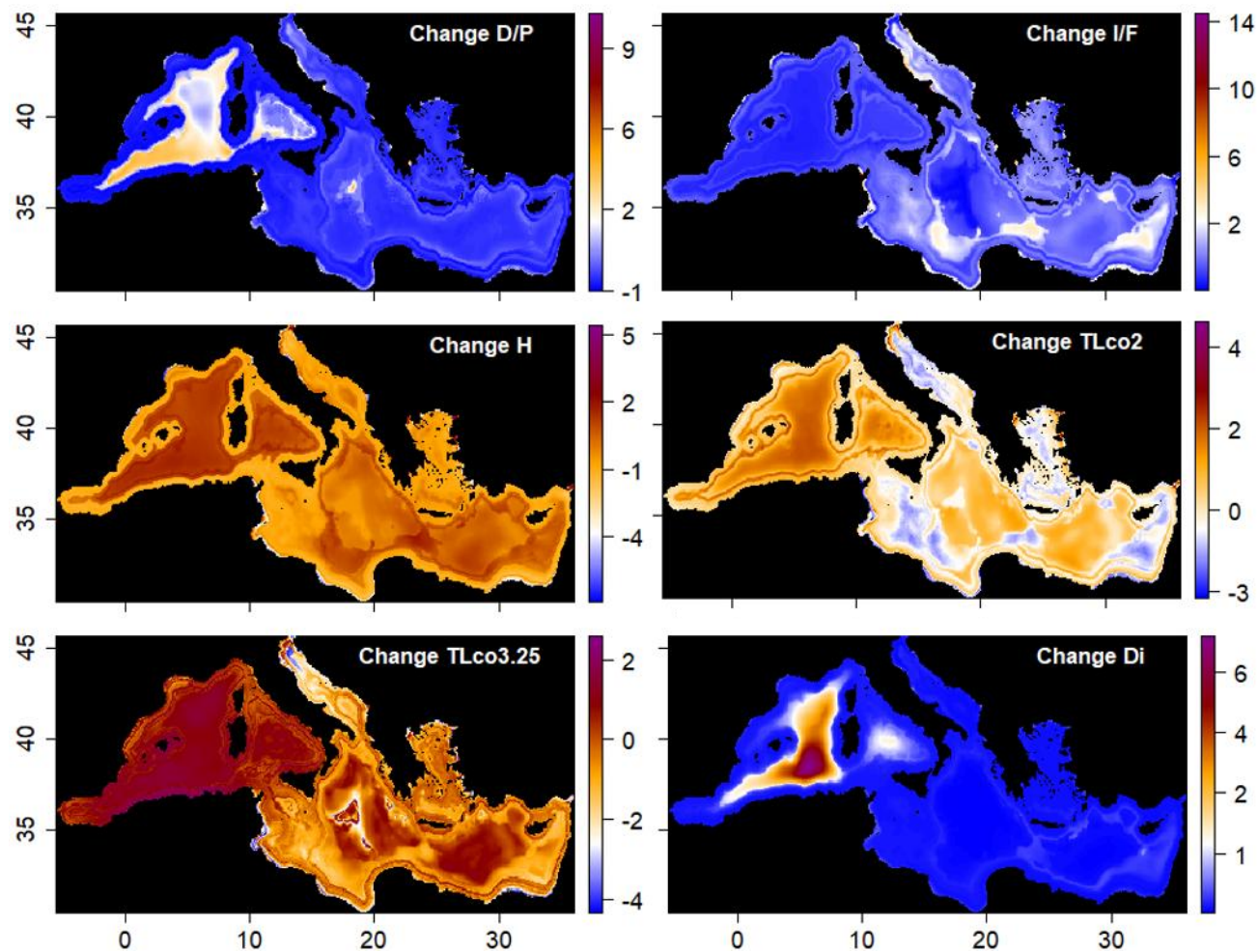


Fig S13. Mean (1995-2016) change (%) for selected modelled indicators. D/P: Demersal/Pelagic fish; I/F: Invertebrates/Fish; H: Shannon index; TLco2: Trophic level (TL) of species/Fg in the community with $TL \geq 2$; Trophic level (TL) of species/Fg in the community with $TL \geq 3.25$; Di: Discards. To facilitate the comparison, indicators were standardized (difference from the mean divided by the corresponding standard deviation).